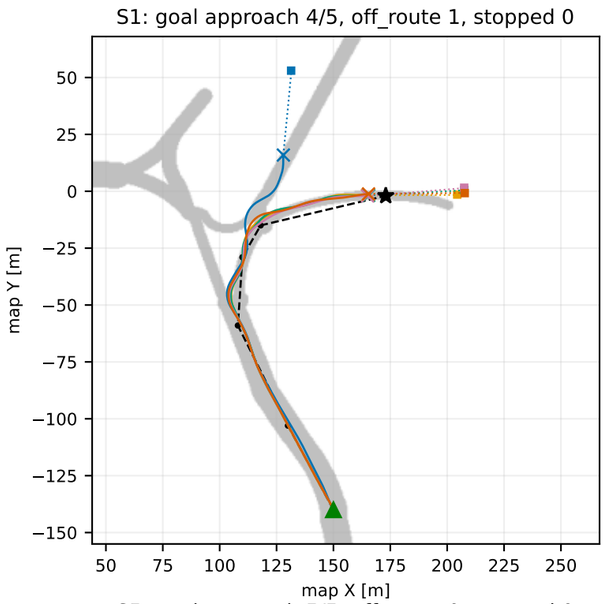
전체 HIL 60회 결과·궤적·헤딩 보고

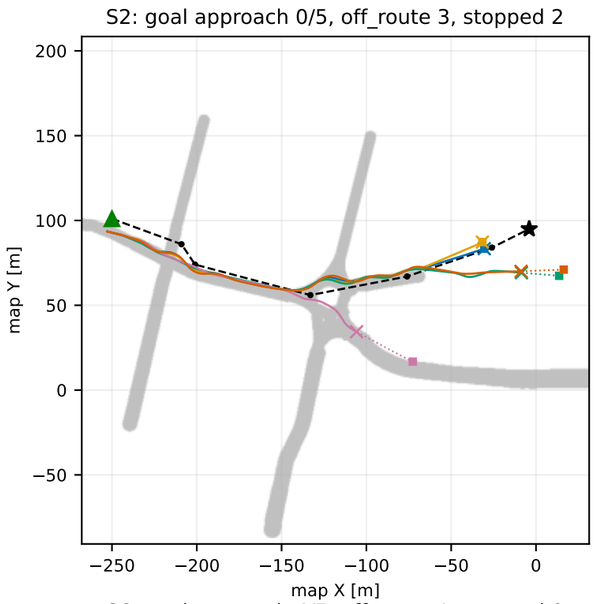
2026-10-09 | 20261009_070714_all_d8c3cf
원래 시나리오 1~12 × 각 5회 · 선택 장면이 아닌 전체 시행

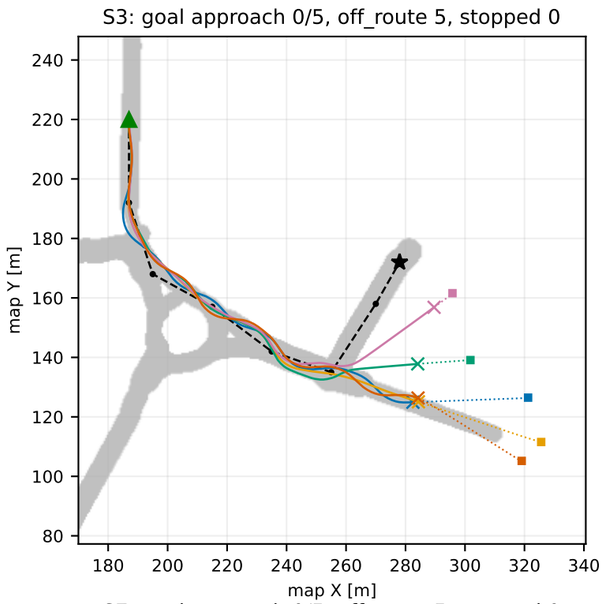
Original S	Goal approach	off_route	stopped_10s	Stop confirmed
1	4	1	0	5
2	0	3	2	5
3	0	5	0	5
4	5	0	0	5
5	5	0	0	5
6	4	1	0	5
7	0	5	0	5
8	0	5	0	5
9	5	0	0	5
10	1	4	0	5
11	1	4	0	5
12	0	5	0	5

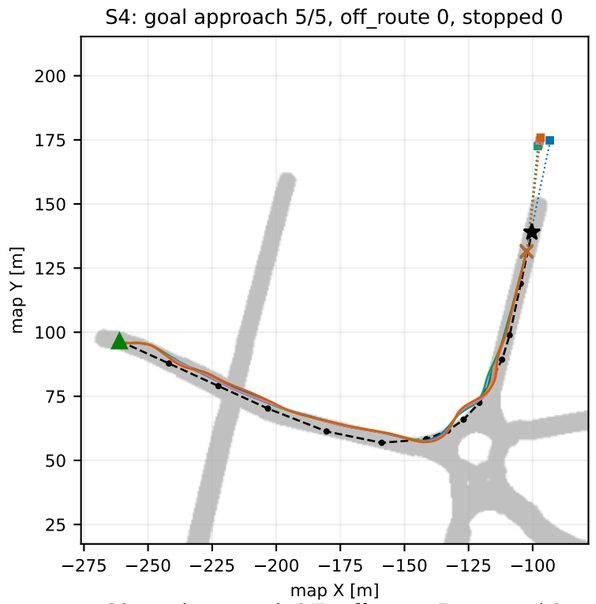
목표 접근 25/60 (41.7%) · off_route 33/60 (55.0%) · stopped_10s 2/60 (3.3%)
60회 모두 출발·정지 확인, 시스템 생략 0회. completed는 목표 정차 성공이 아닙니다.
궤적: running 실선 / 관성 이동 점선. 배경은 설계 지도 재구성입니다.
헤딩: map yaw와 하위 경로 오차를 구분하며 EKF-SIH 차이를 나타내지 않습니다.
원인 분석·튜닝 없음. 보고 번호 1~7의 선택 자료와 구분하여 읽어 주세요.

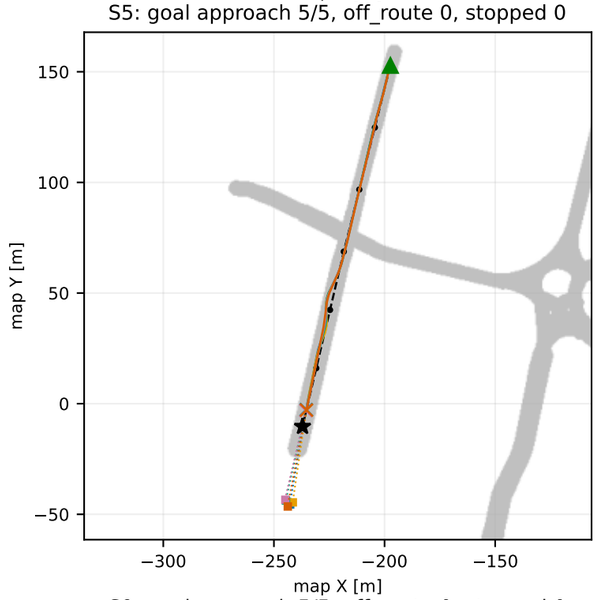
20261009_070714_all_d8c3cf | all 60 attempts | solid=running, dotted=coasting, x=outcome, square=stop
Road backdrop: reconstructed design map, not independent road-containment truth

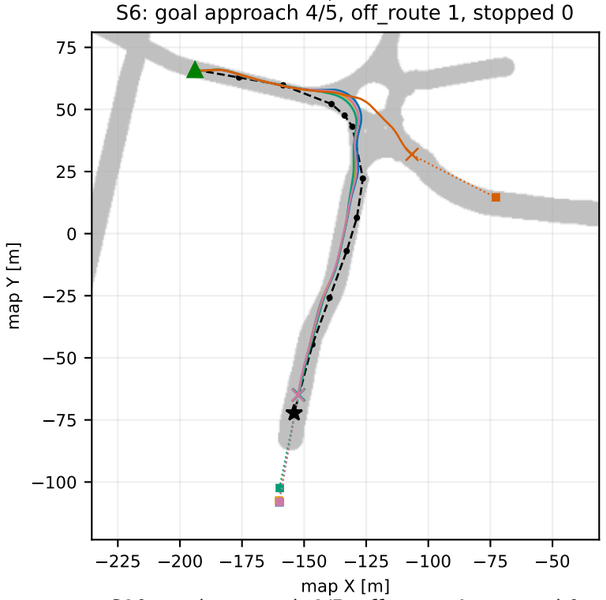


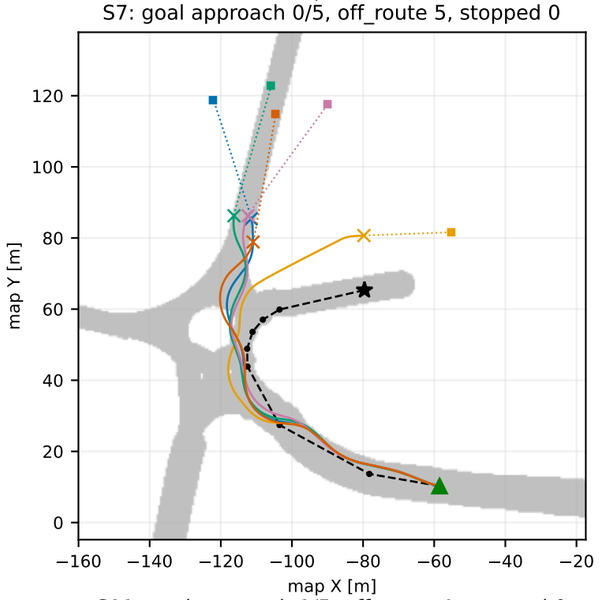


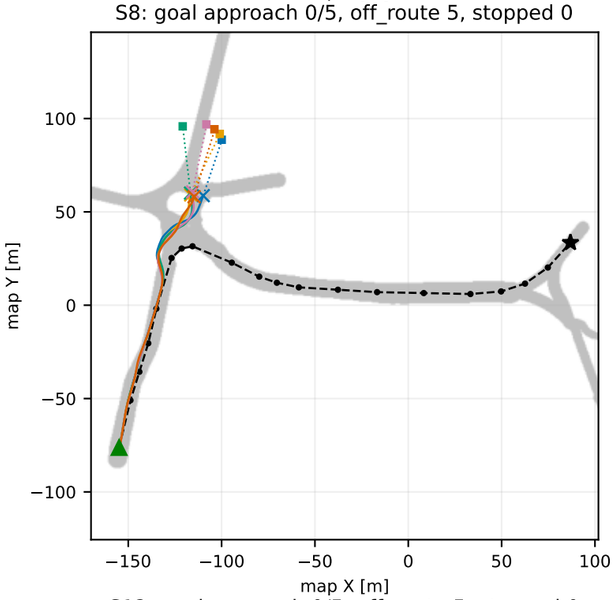


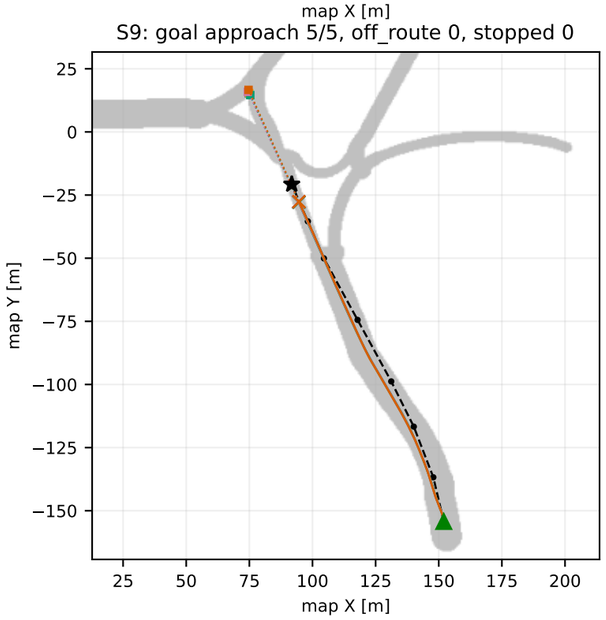


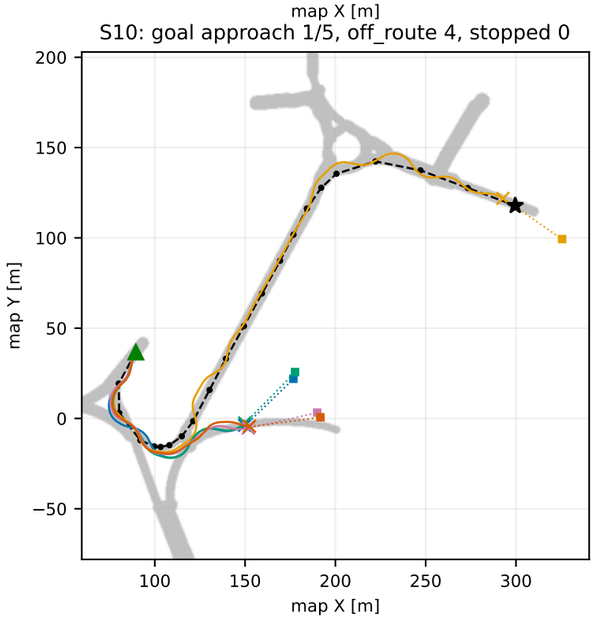


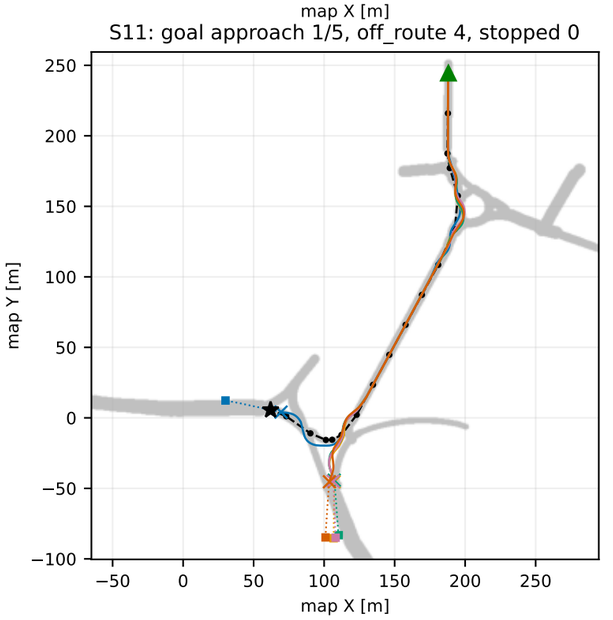


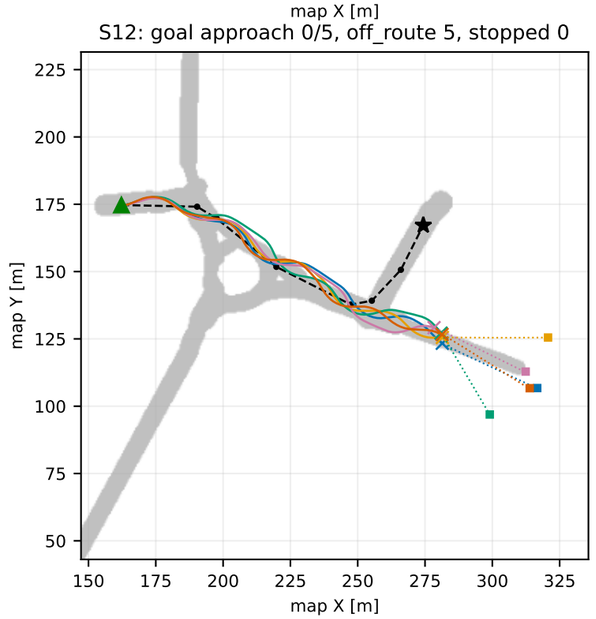




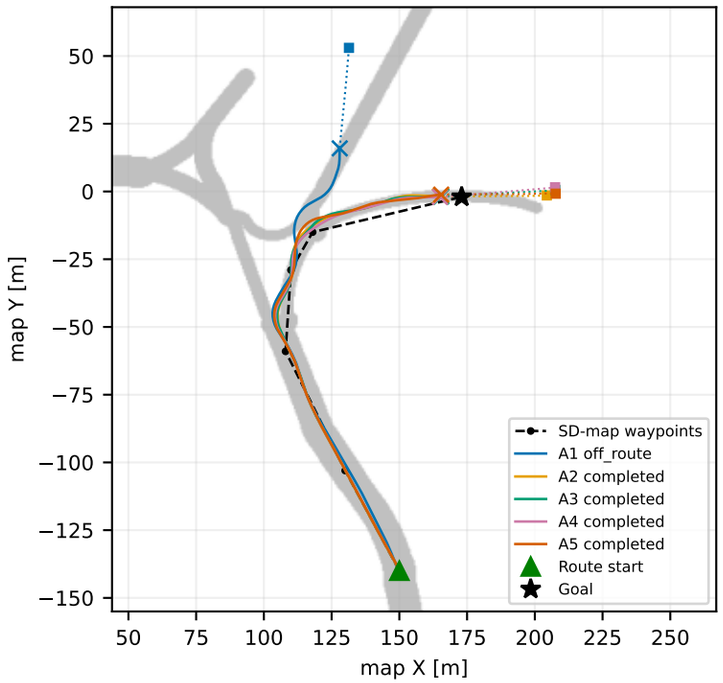


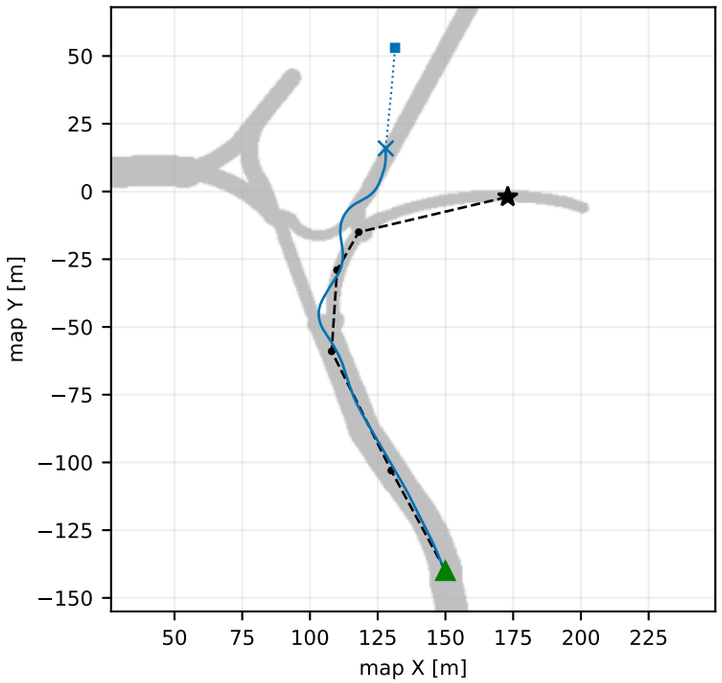


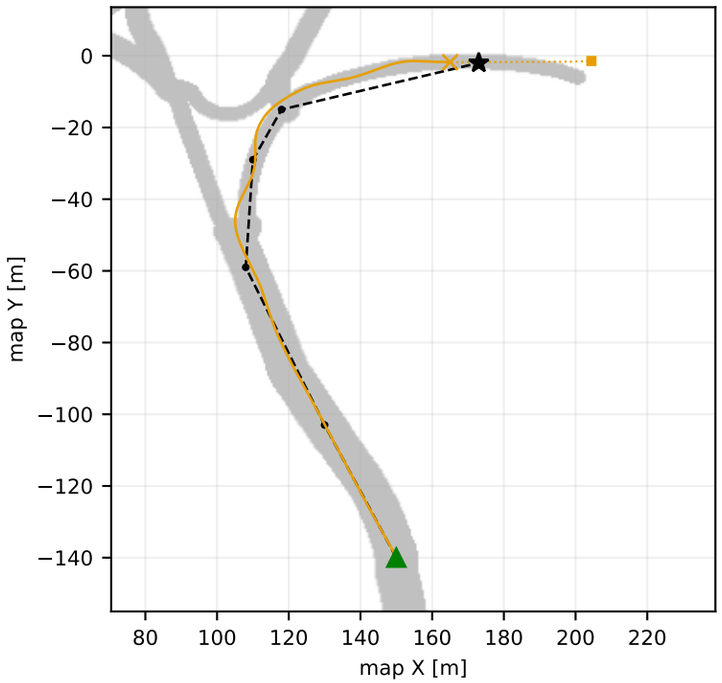


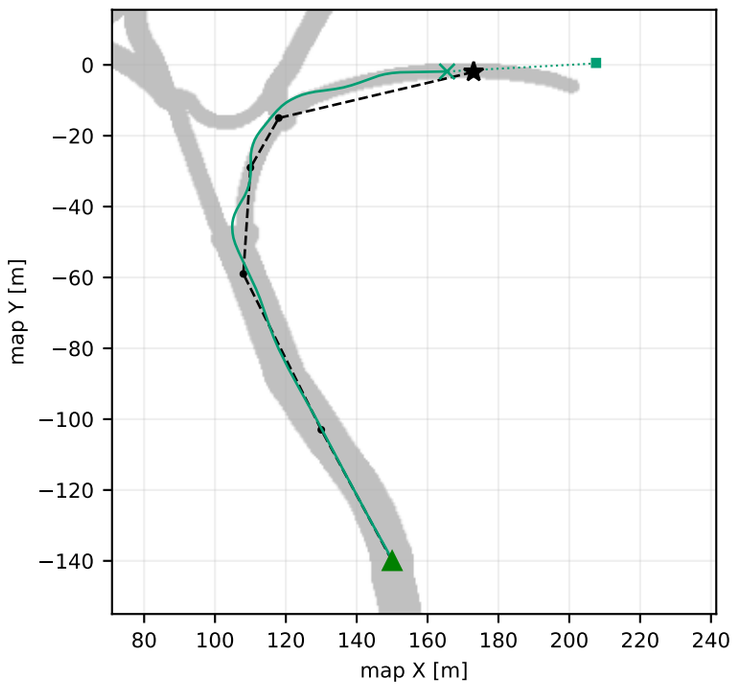


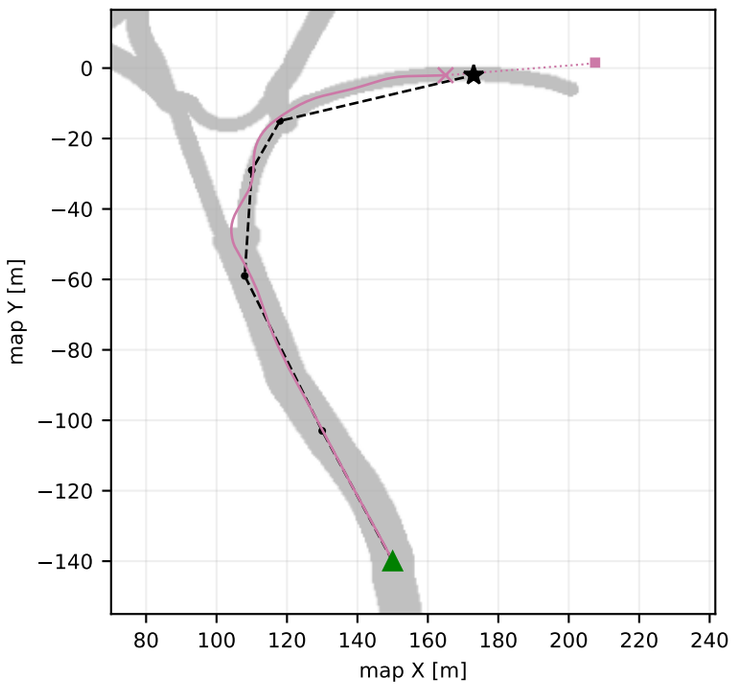
20261009_070714_all_d8c3cf | solid=running, dotted=coast; x=outcome, square=stop | design road backdrop

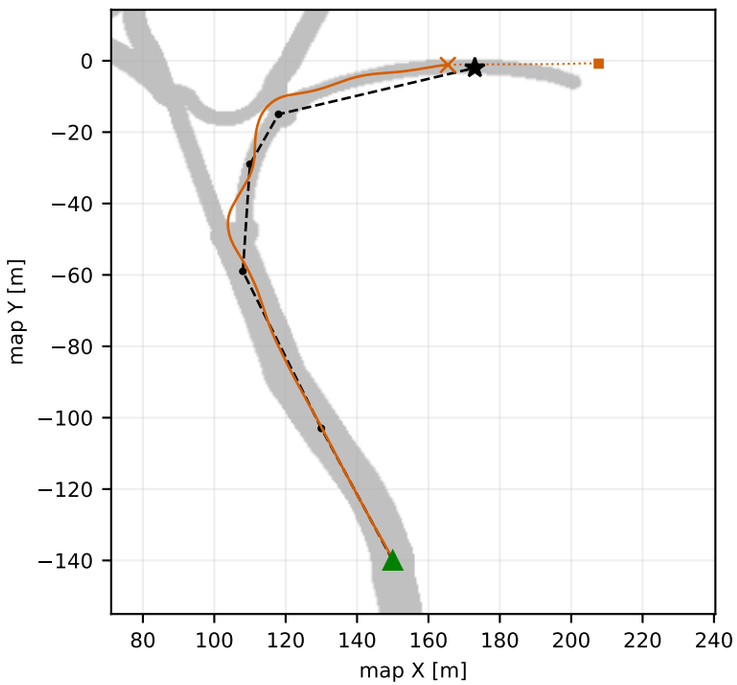




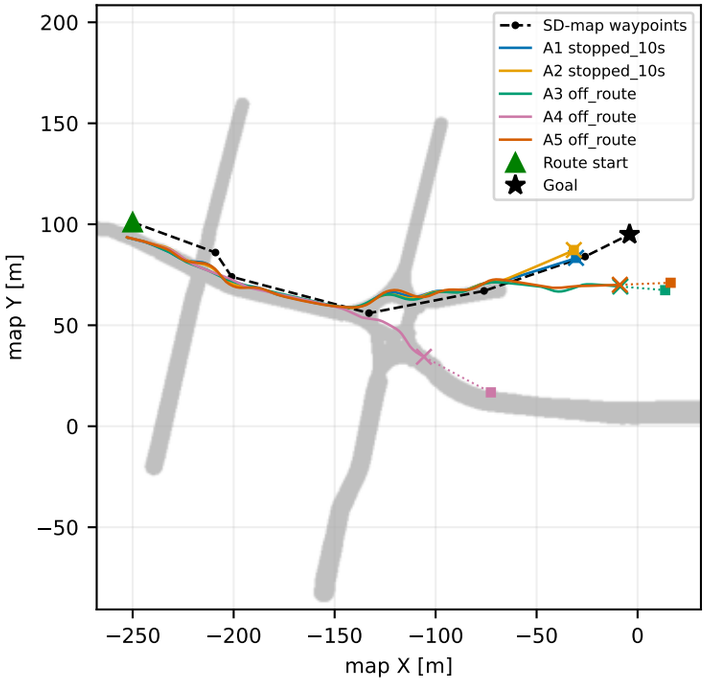


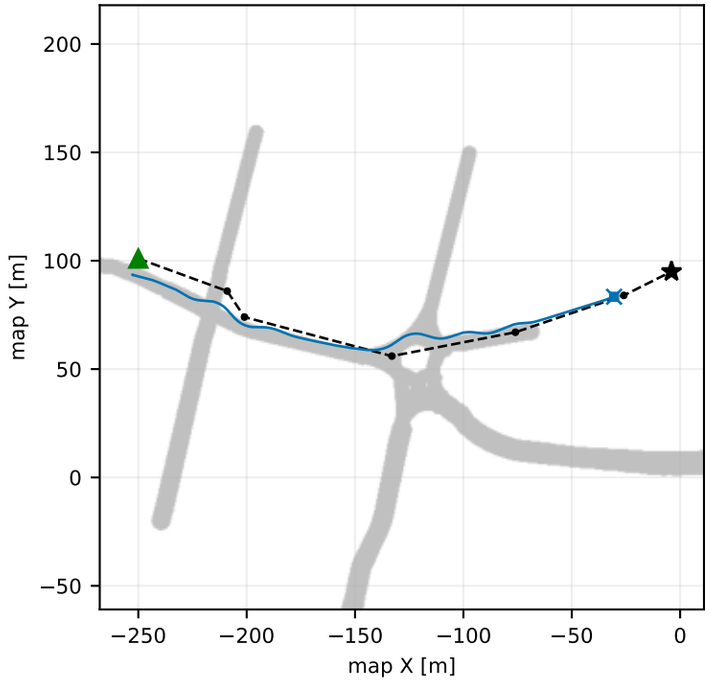


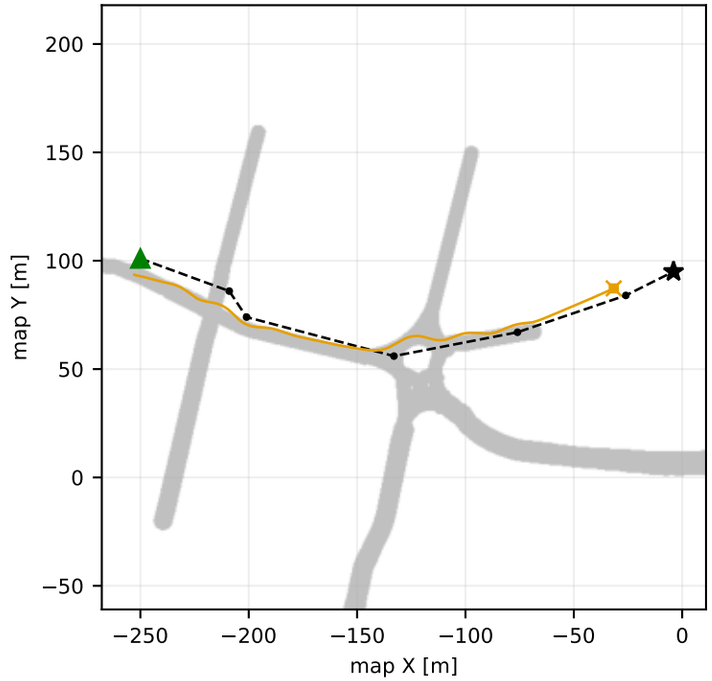


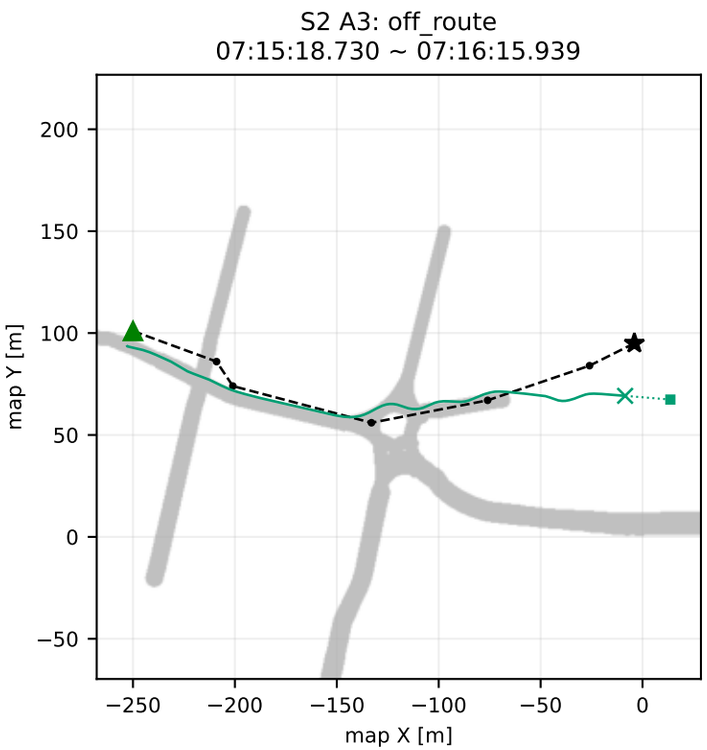


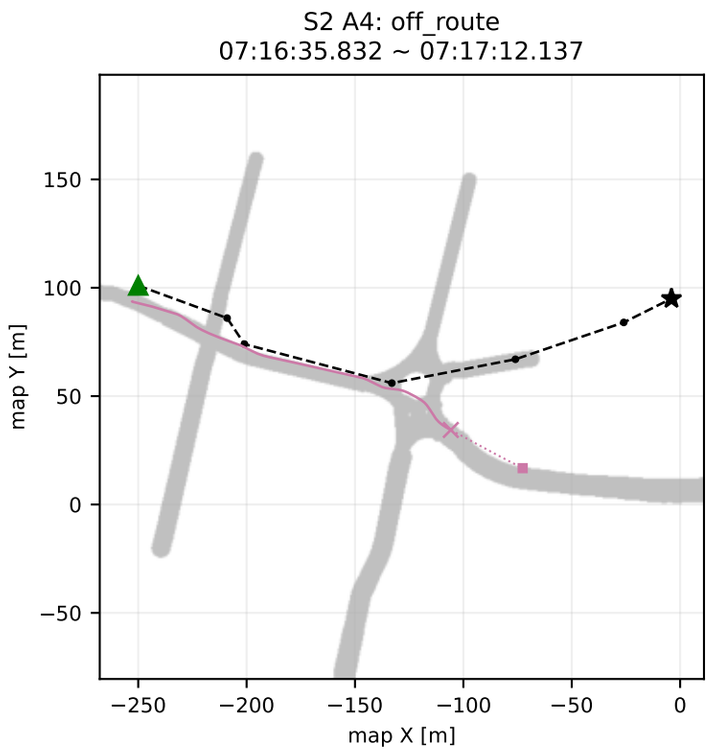
20261009_070714_all_d8c3cf | solid=running, dotted=coast; x=outcome, square=stop | design road backdrop

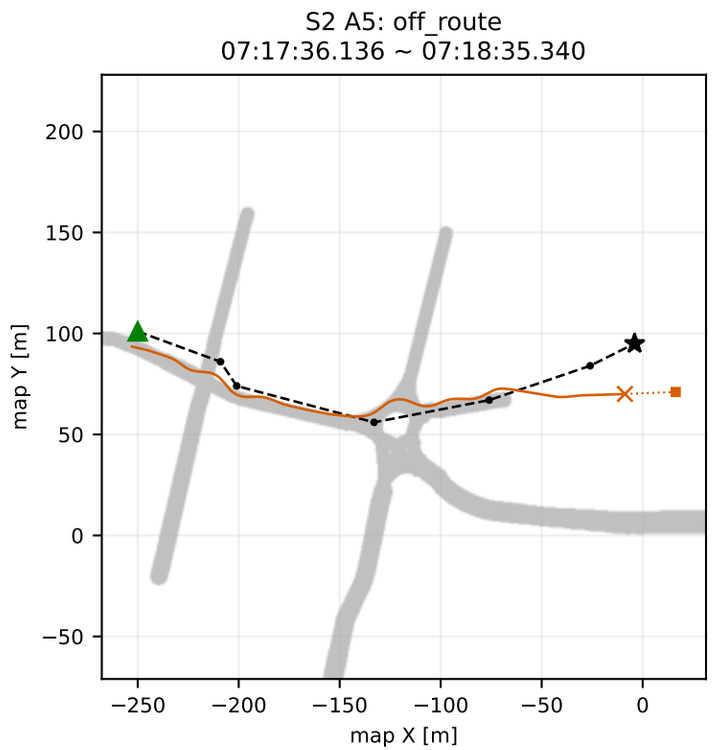




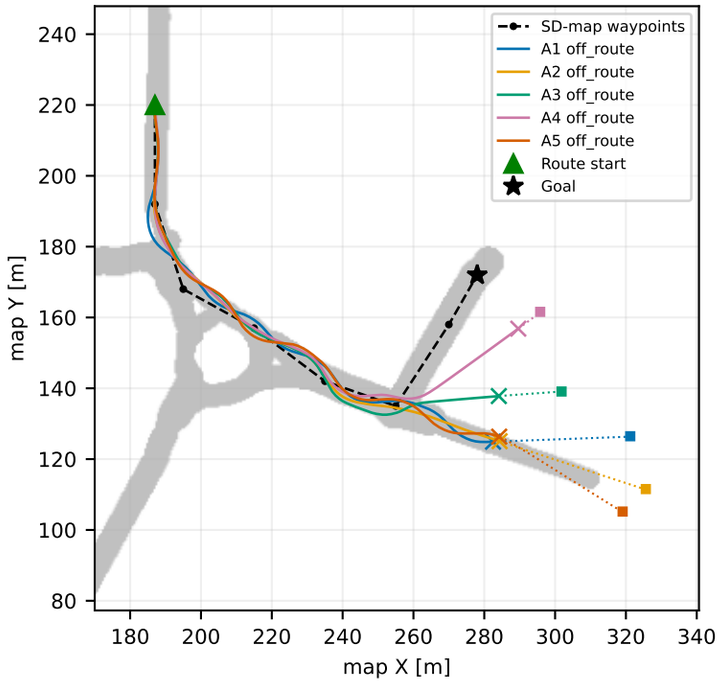


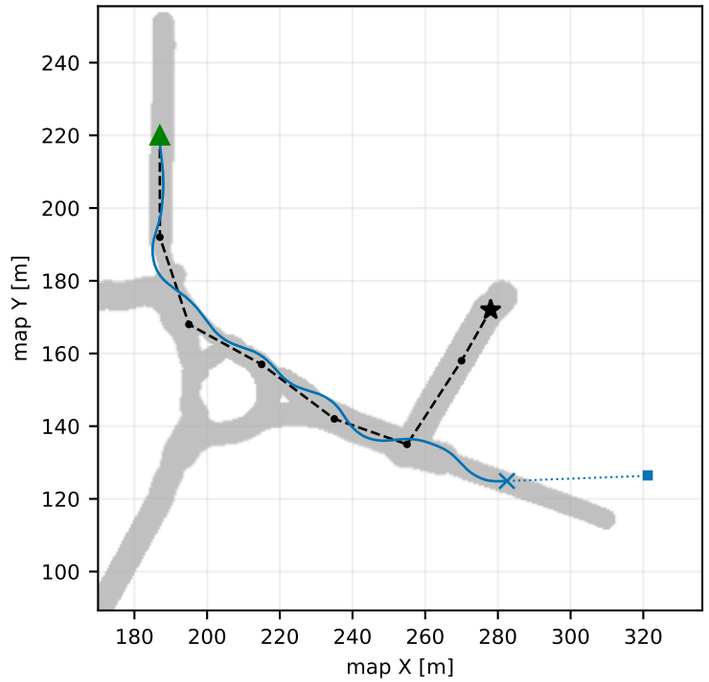


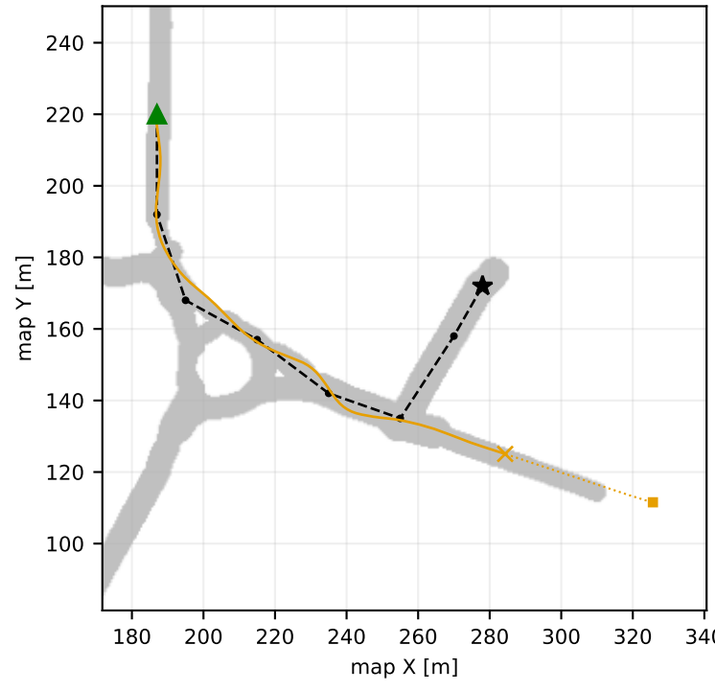


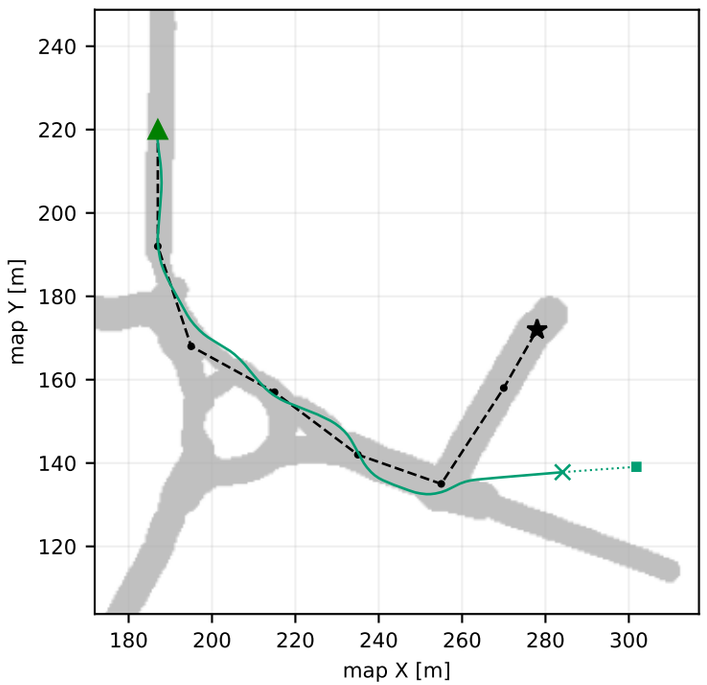


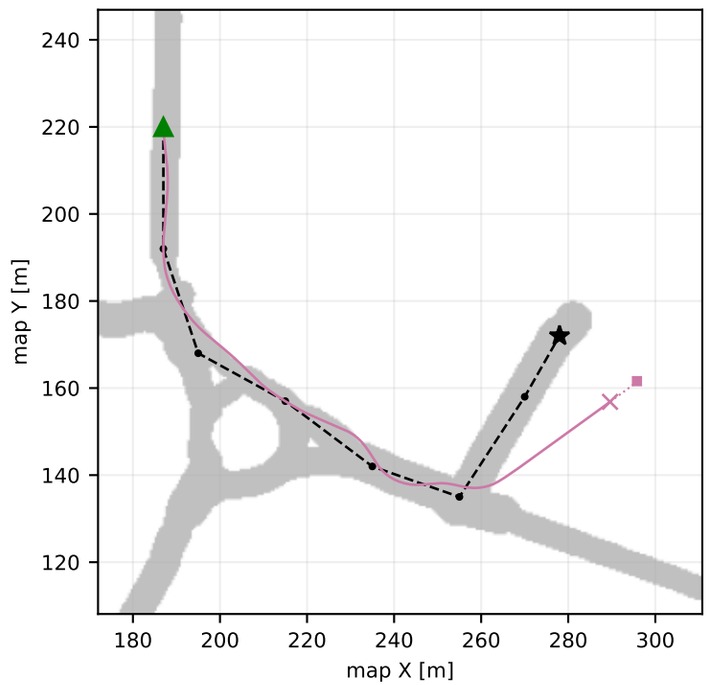
20261009_070714_all_d8c3cf | solid=running, dotted=coast; x=outcome, square=stop | design road backdrop

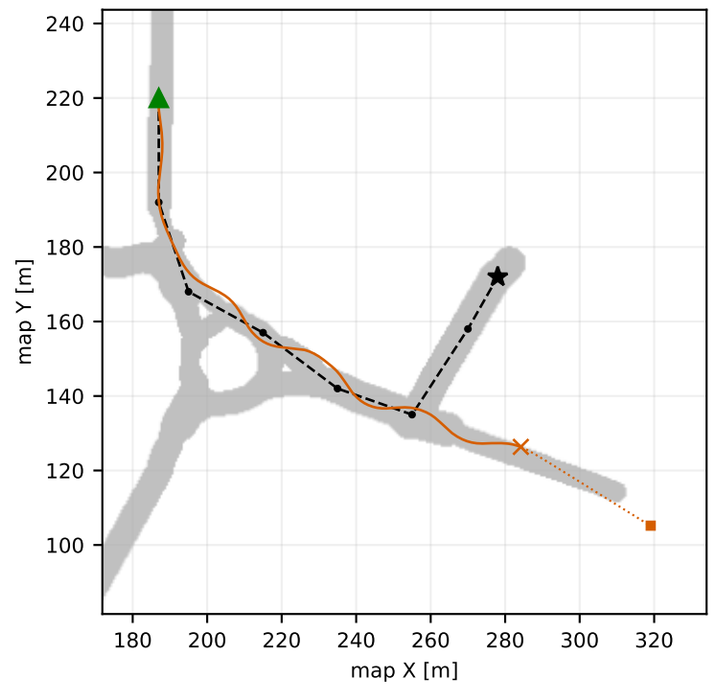




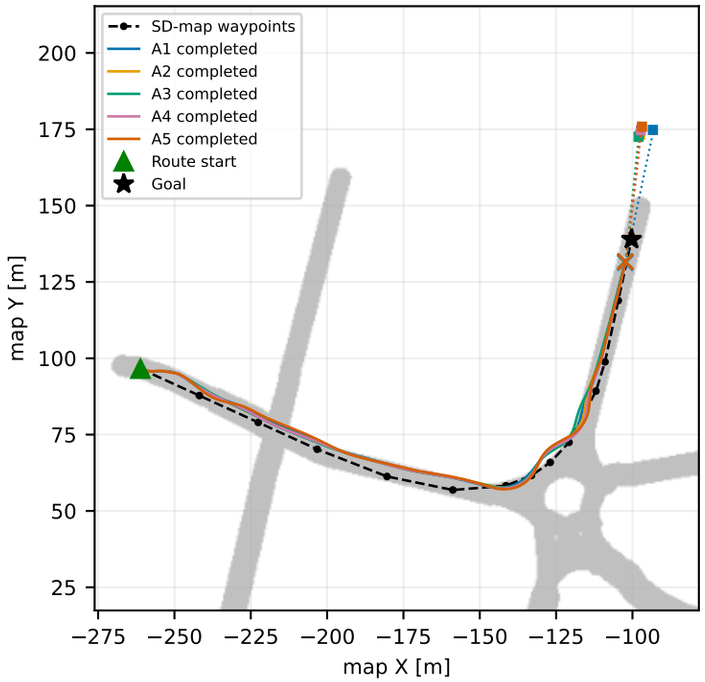


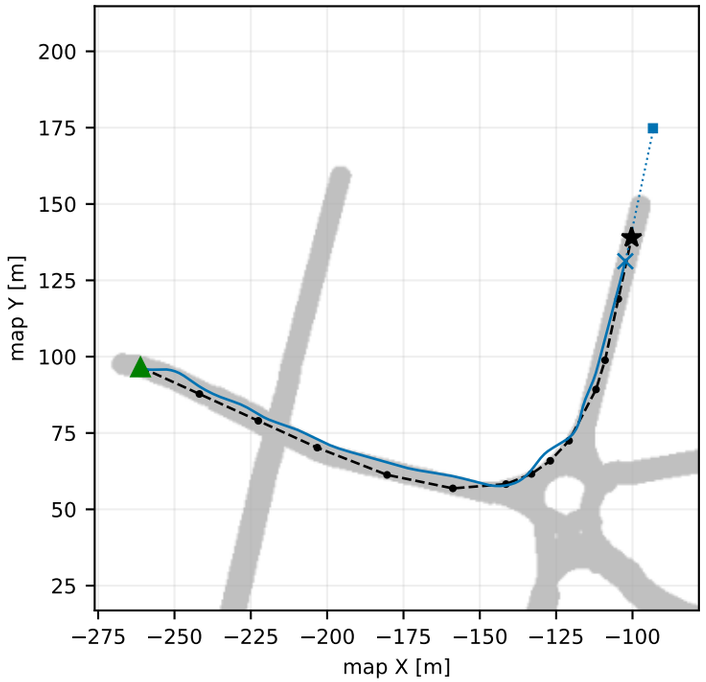


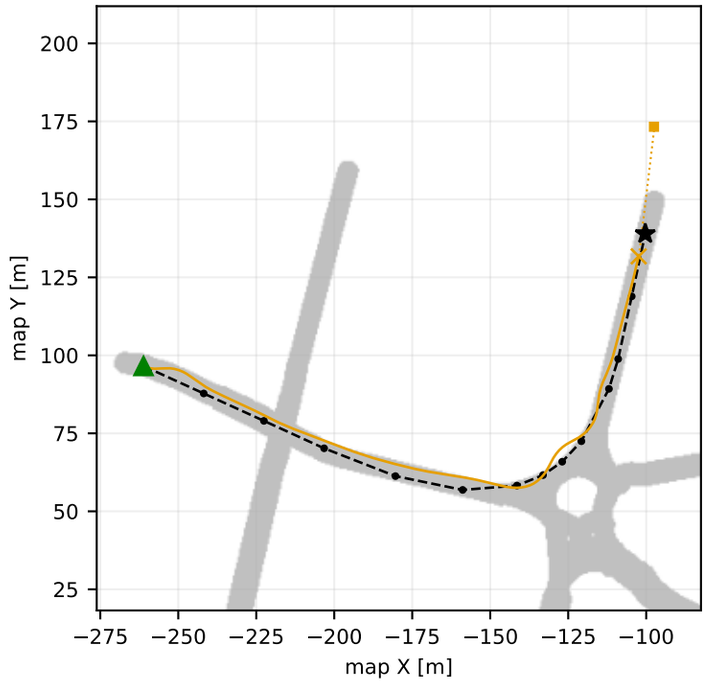


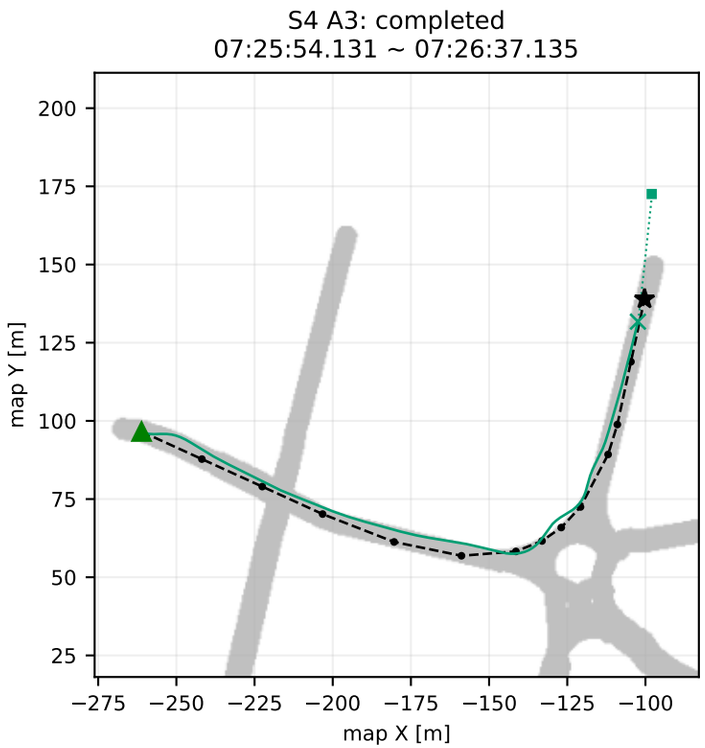


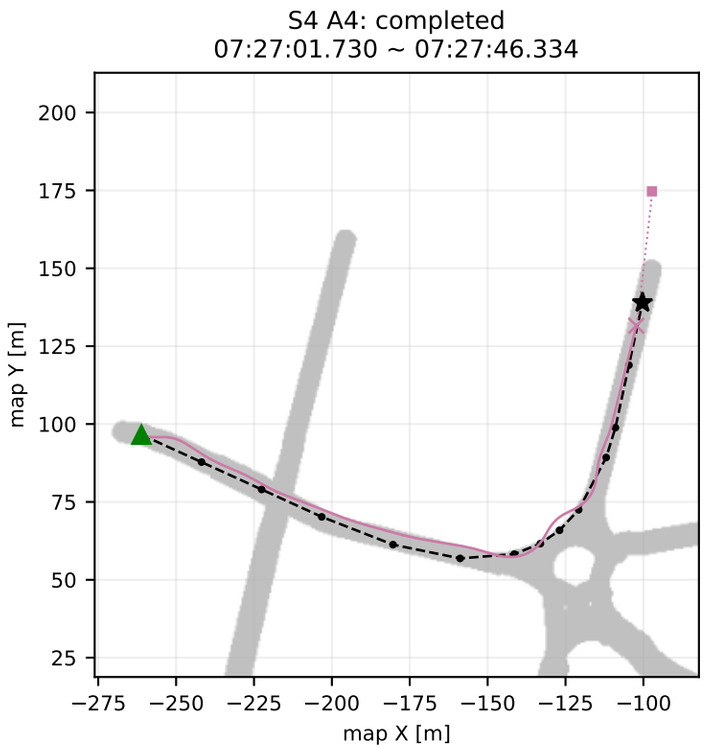
20261009_070714_all_d8c3cf | solid=running, dotted=coast; x=outcome, square=stop | design road backdrop

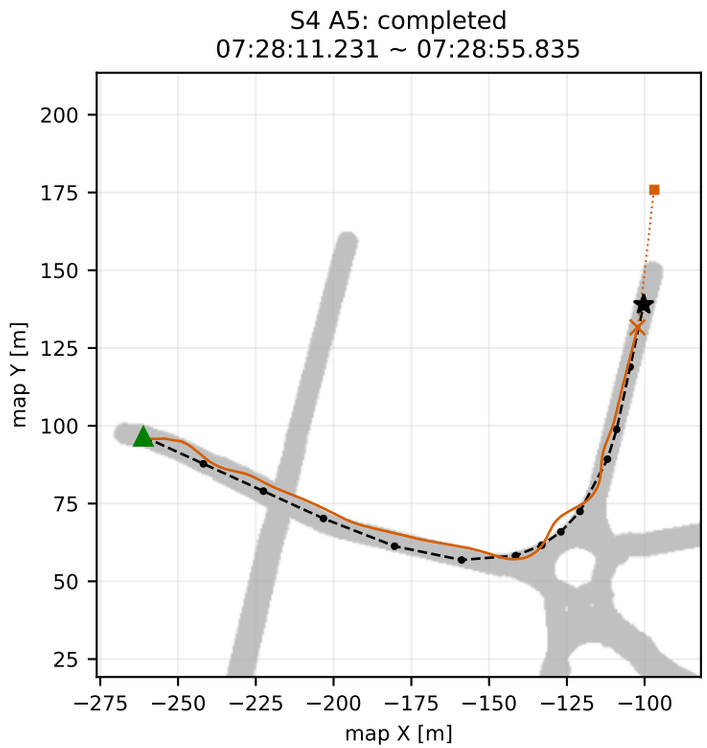




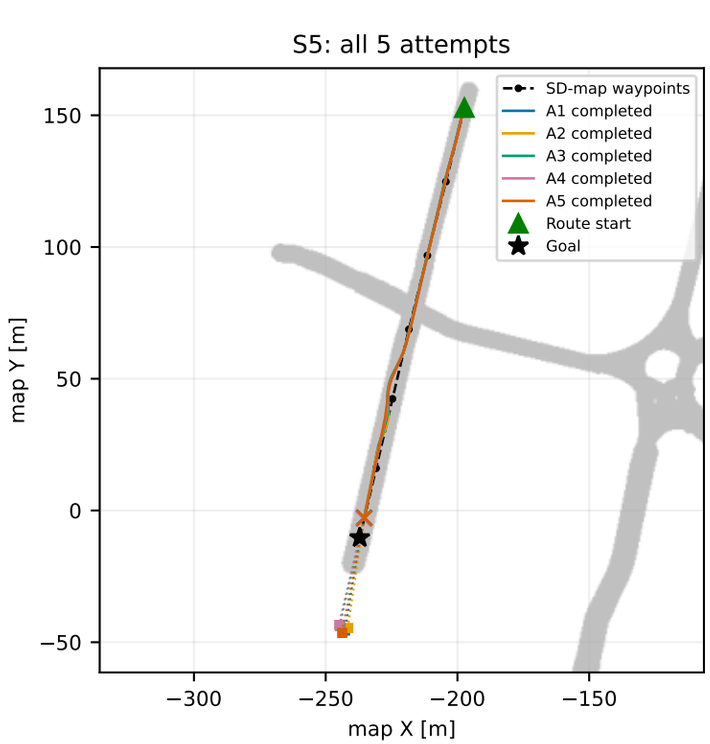


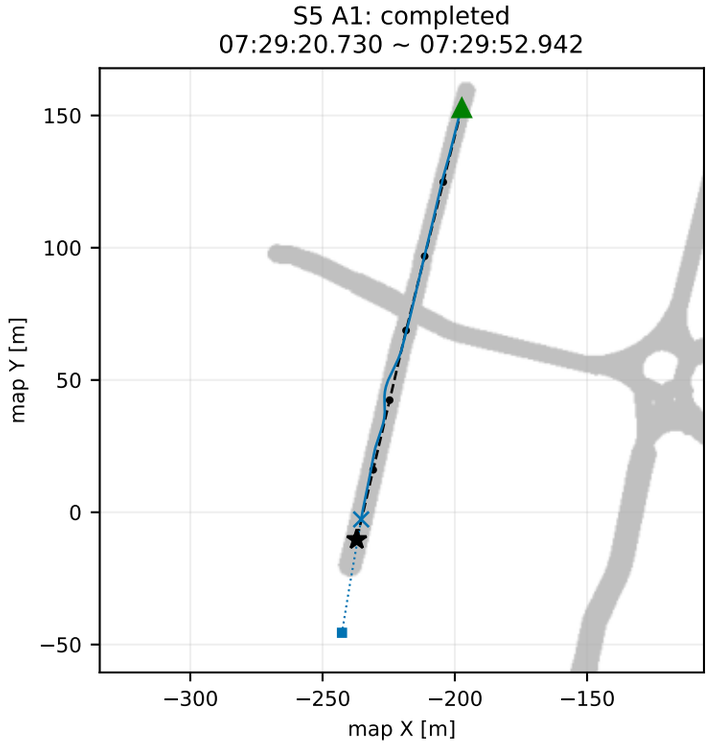


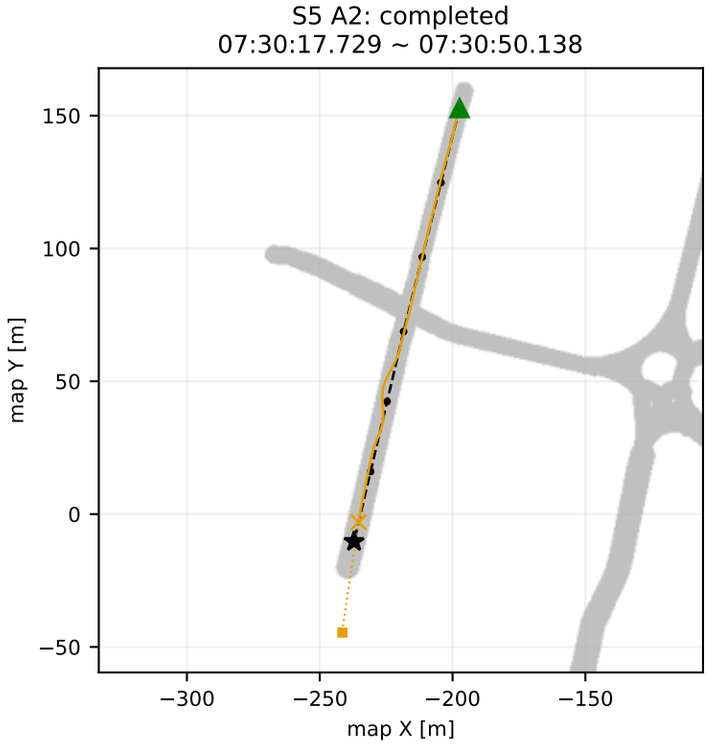


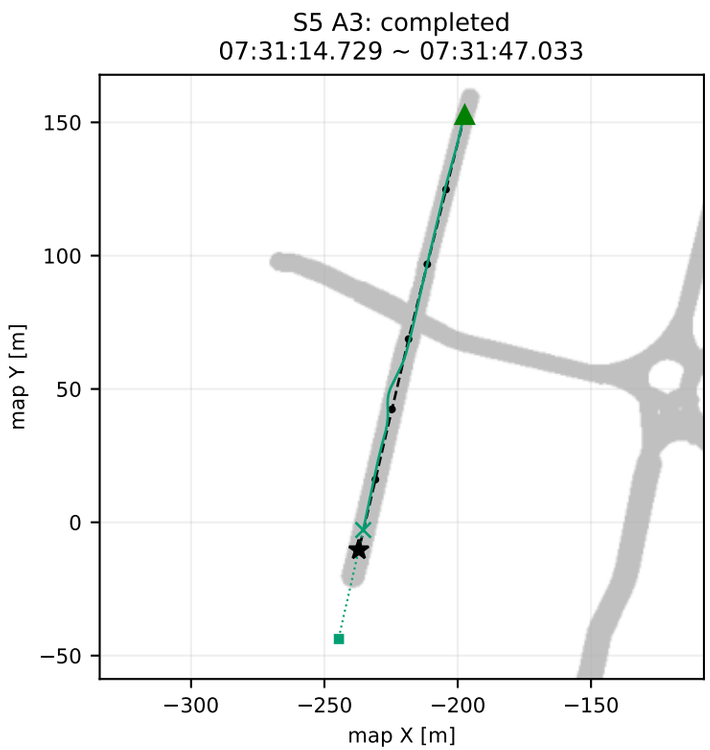


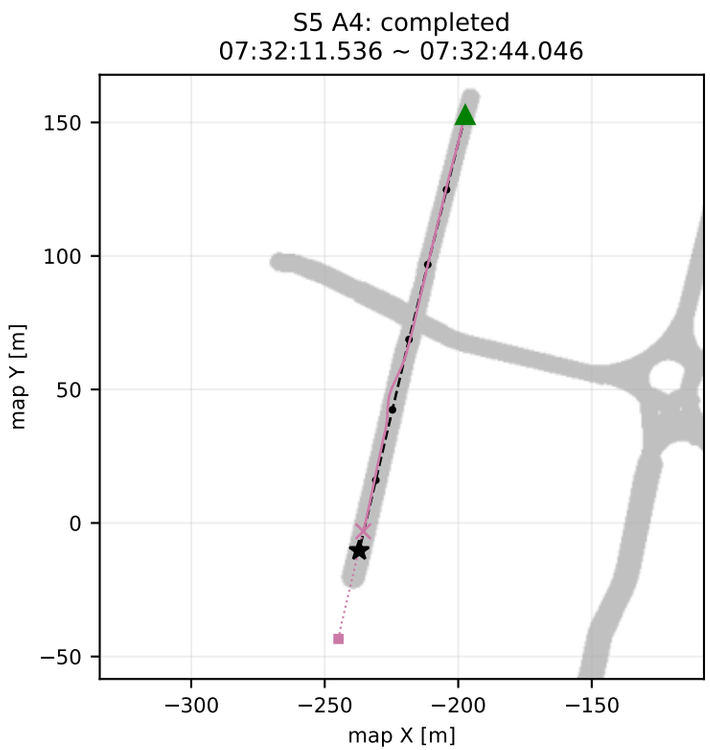
20261009_070714_all_d8c3cf | solid=running, dotted=coast; x=outcome, square=stop | design road backdrop

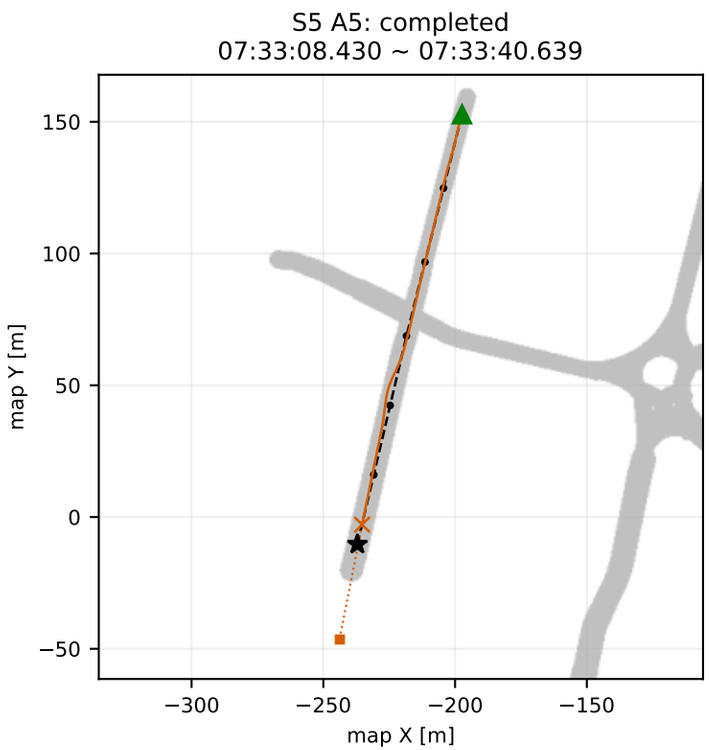




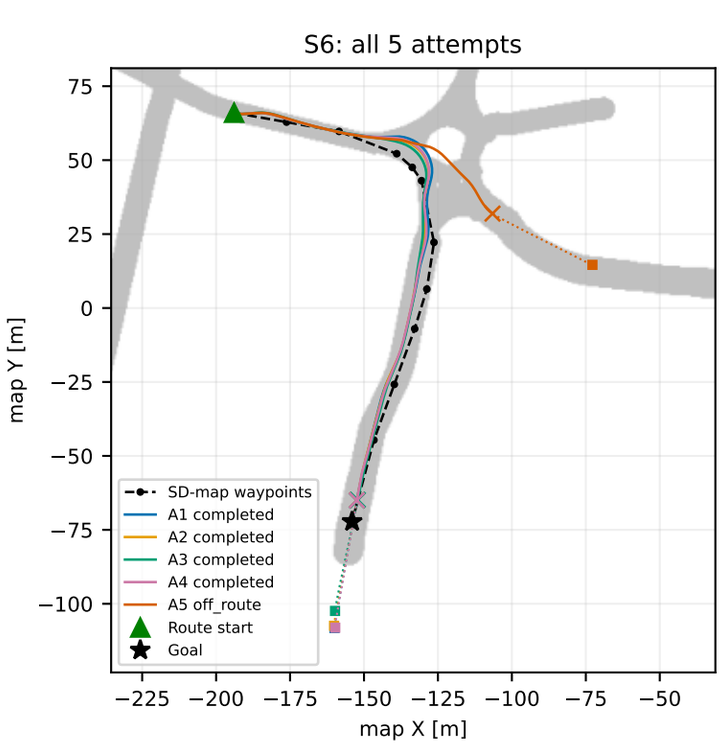


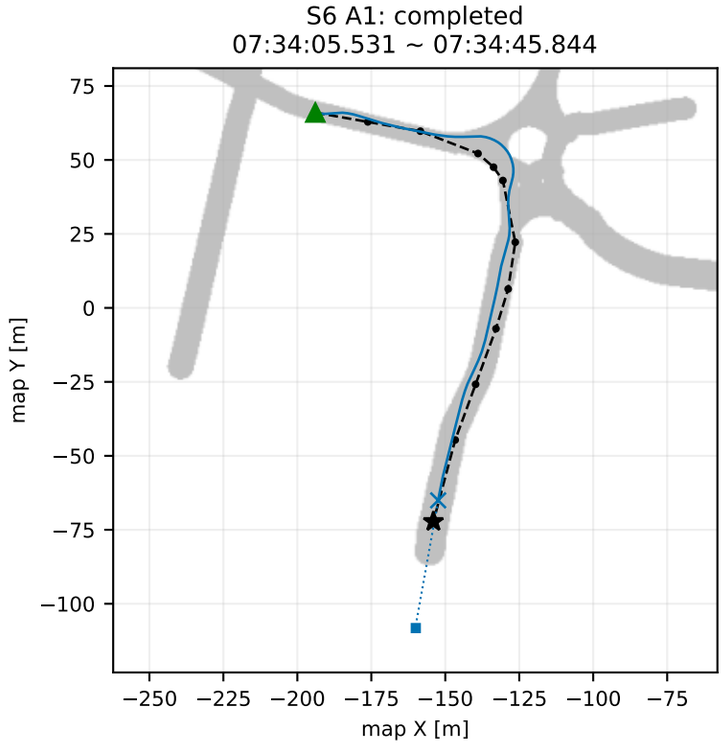


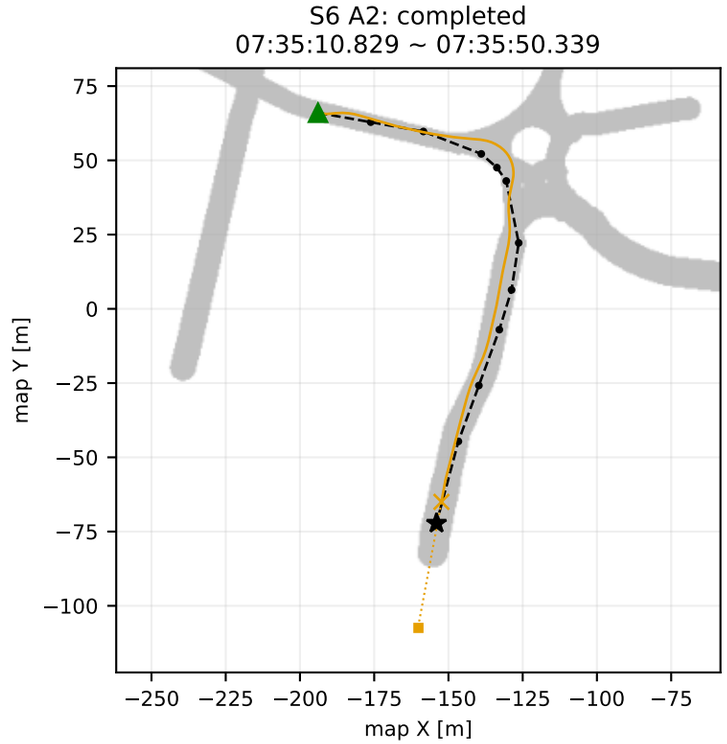


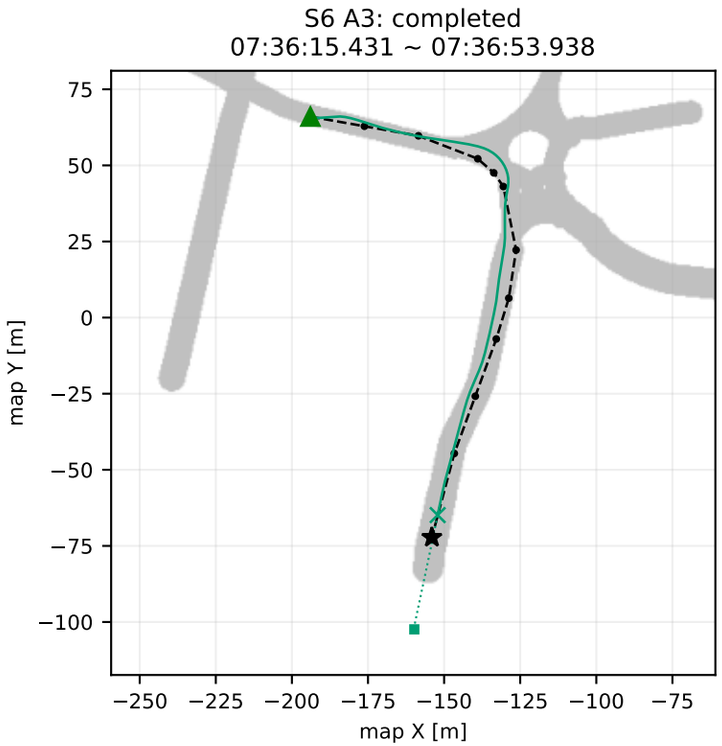


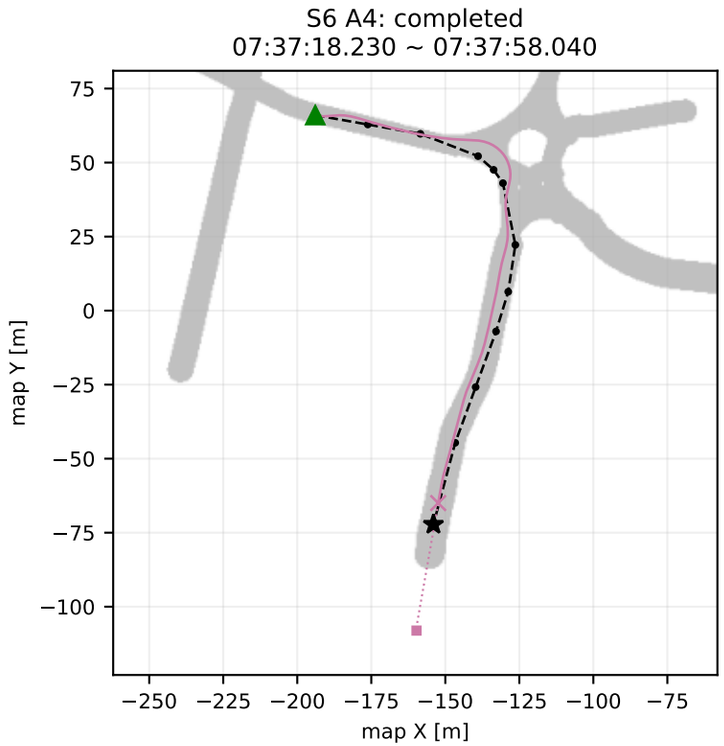
20261009_070714_all_d8c3cf | solid=running, dotted=coast; x=outcome, square=stop | design road backdrop

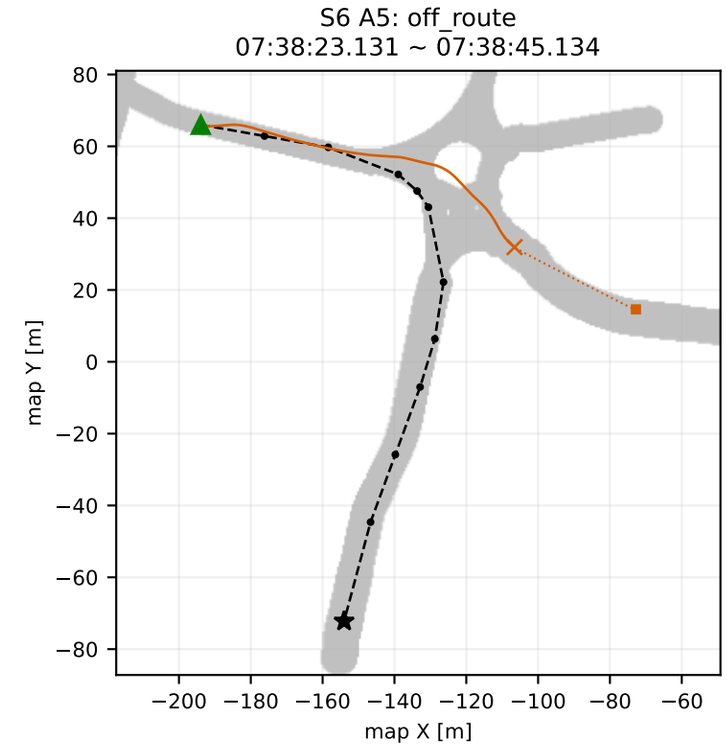




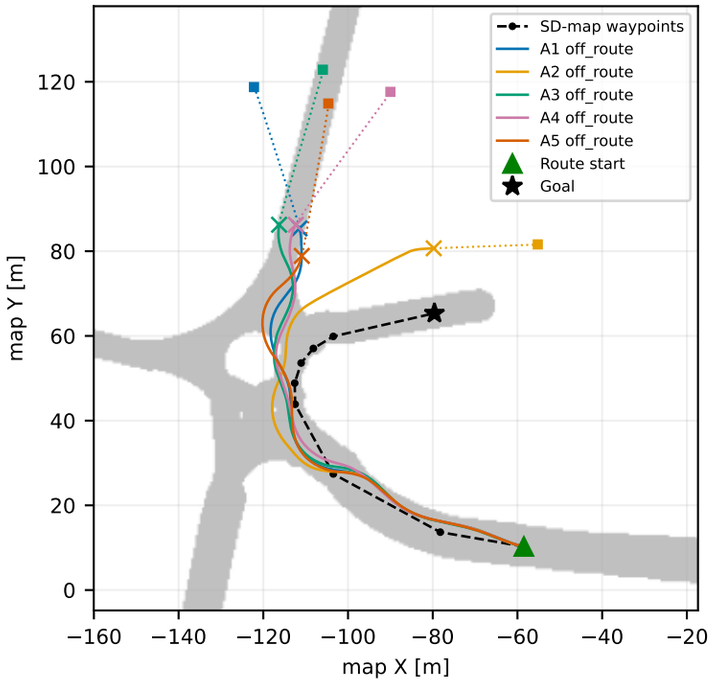


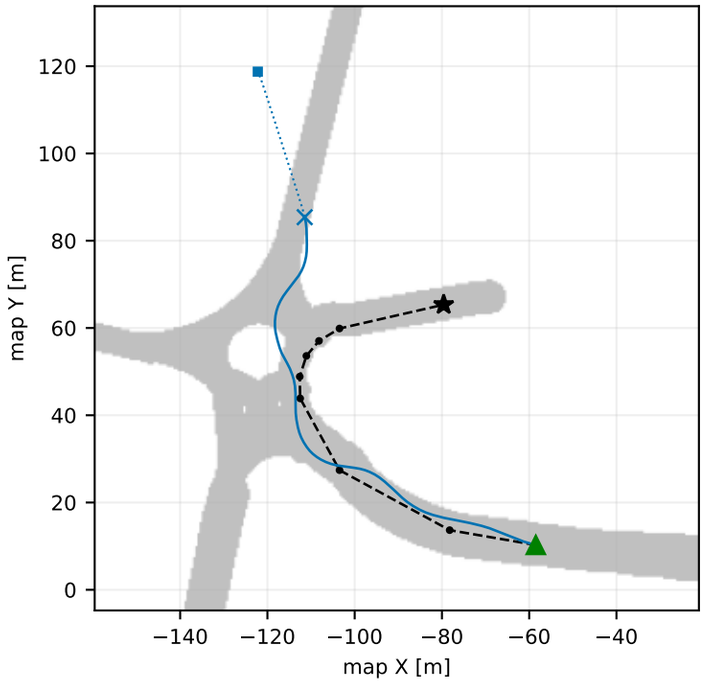


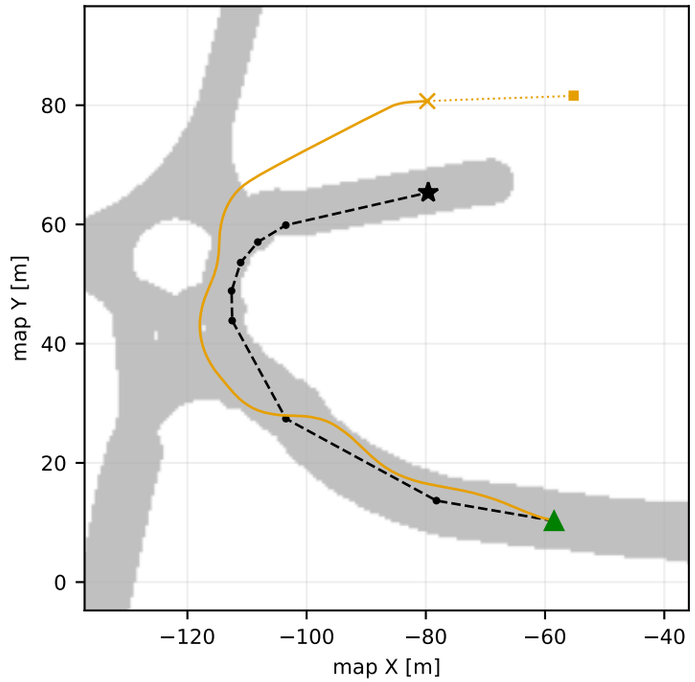


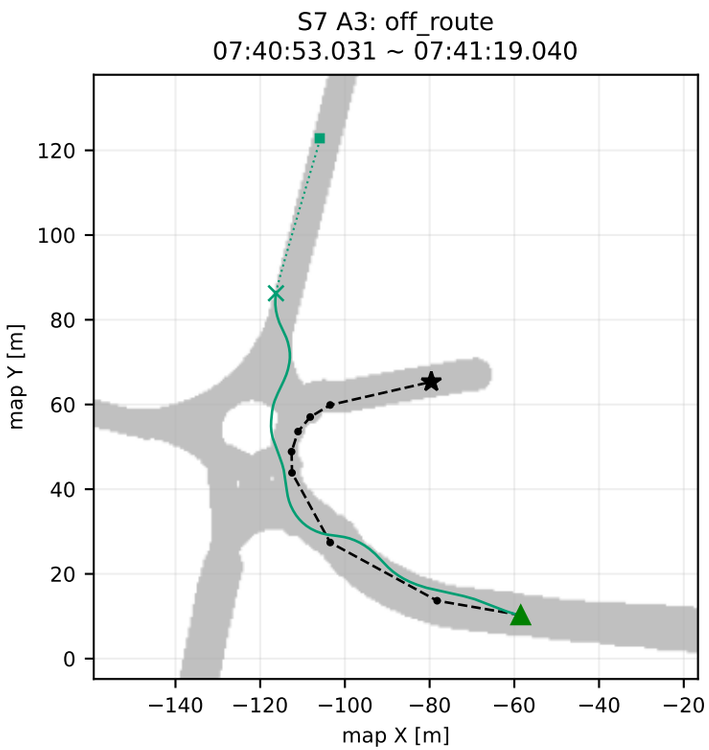


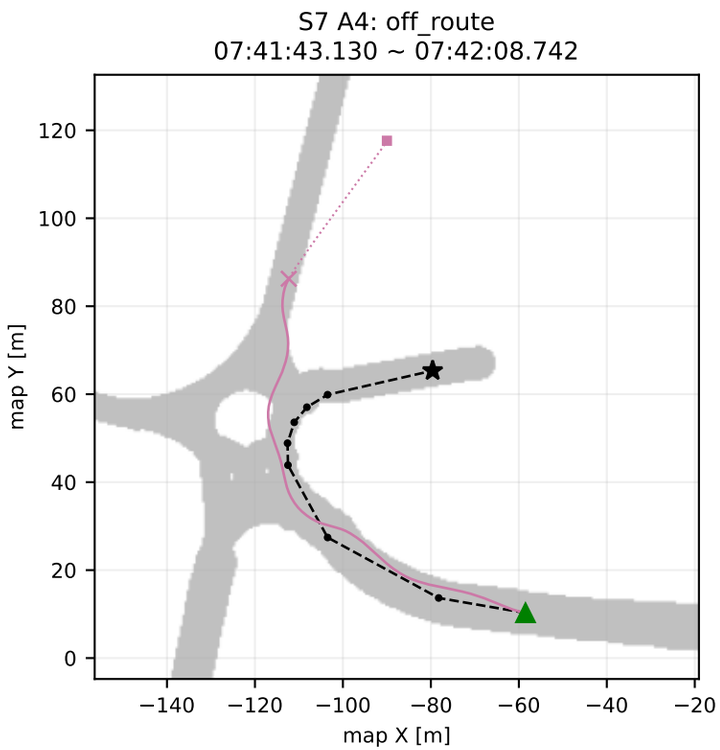
20261009_070714_all_d8c3cf | solid=running, dotted=coast; x=outcome, square=stop | design road backdrop

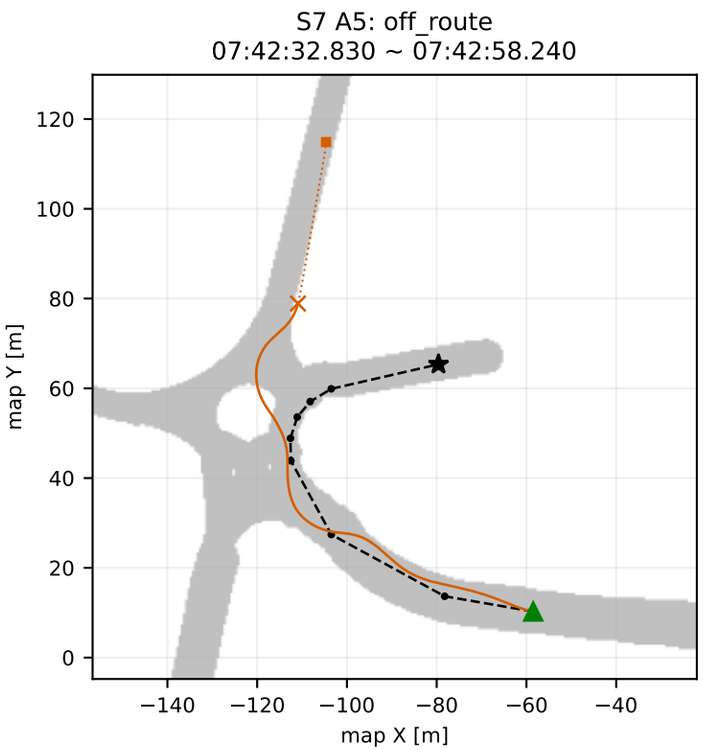




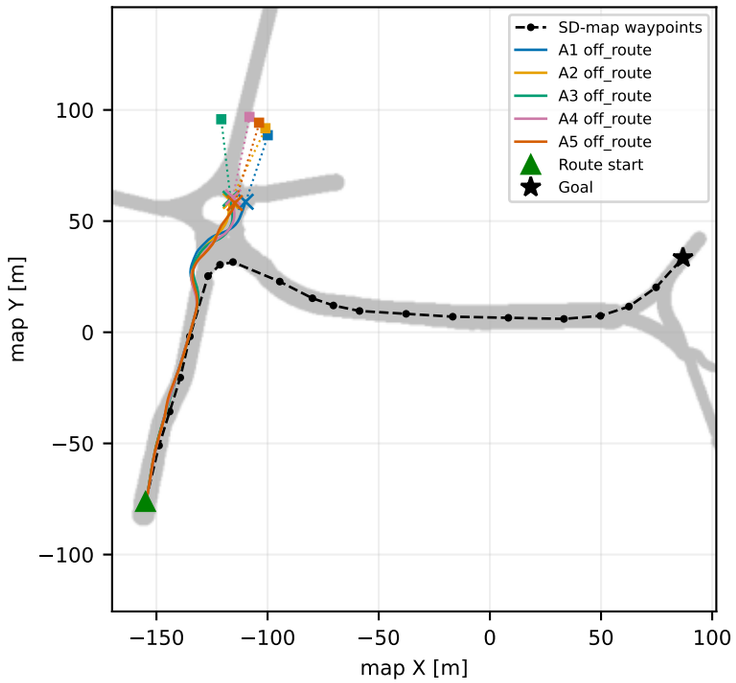


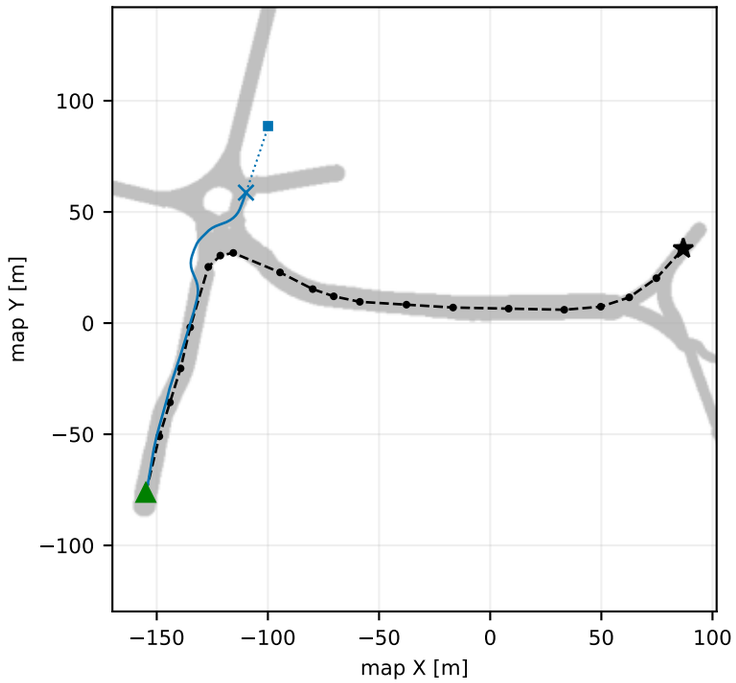


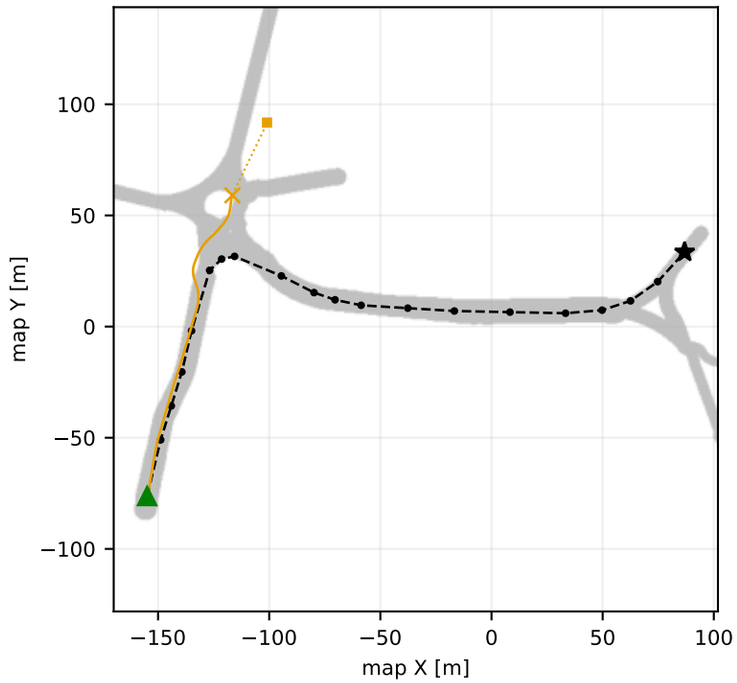


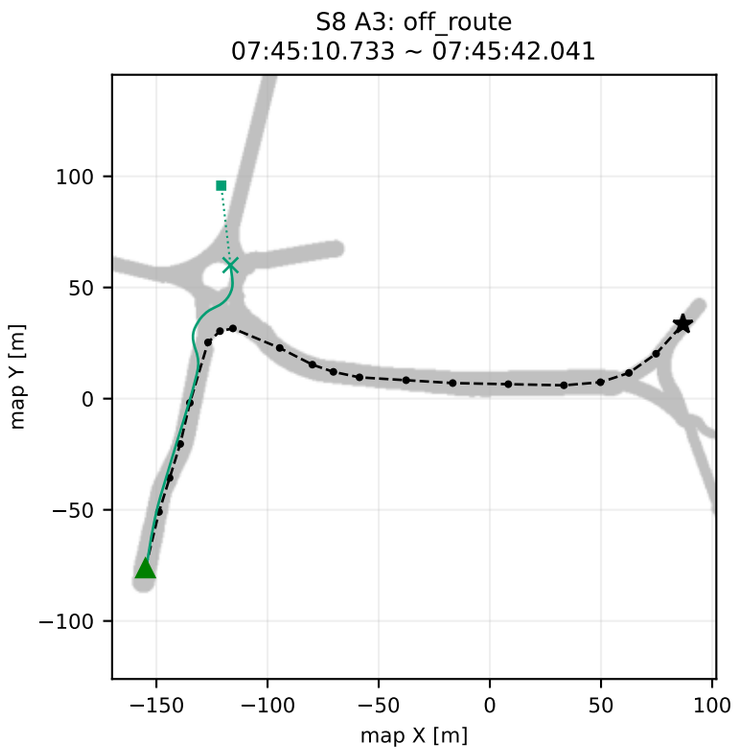


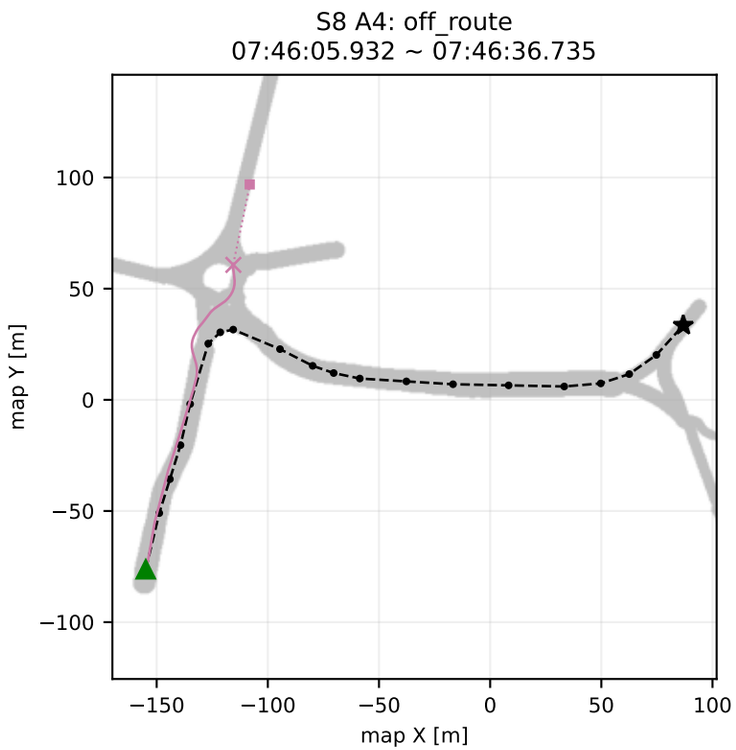
20261009_070714_all_d8c3cf | solid=running, dotted=coast; x=outcome, square=stop | design road backdrop

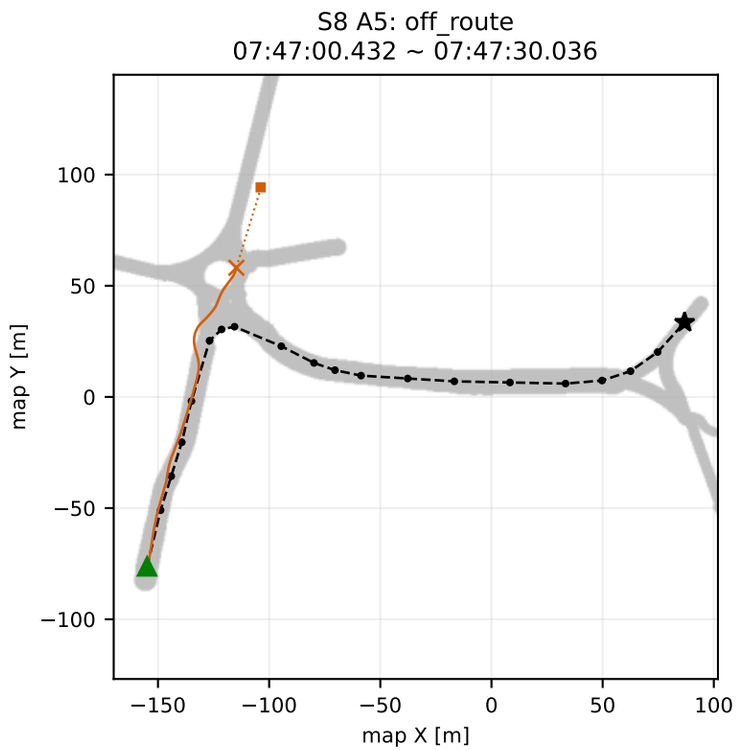




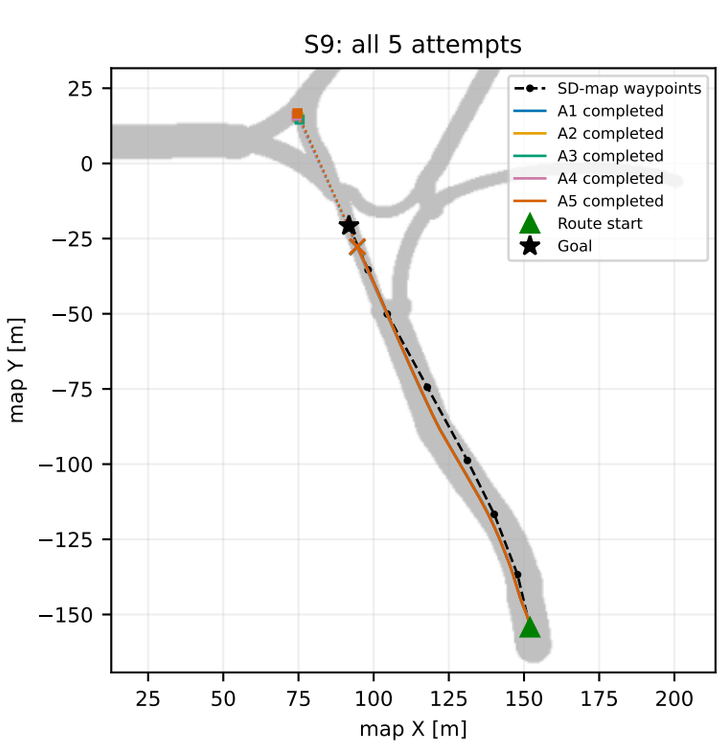


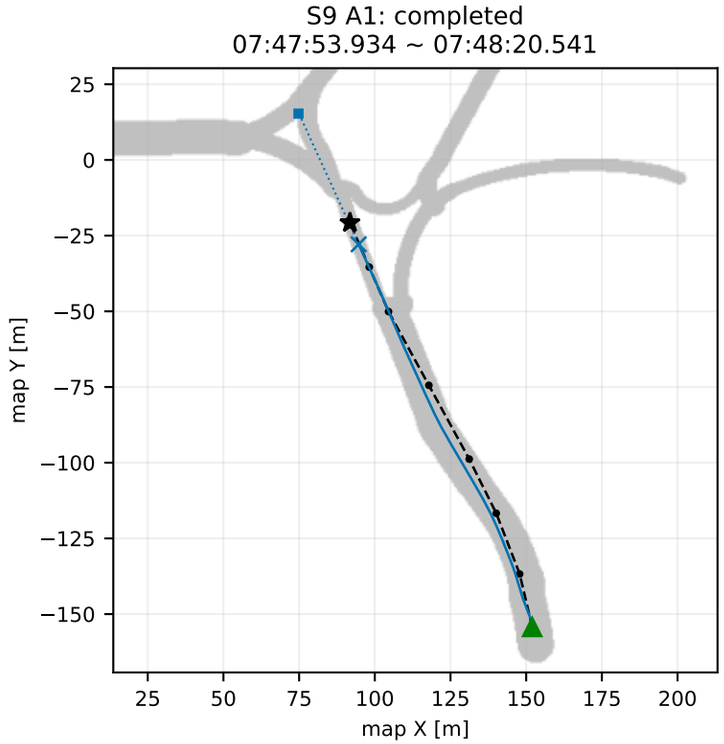


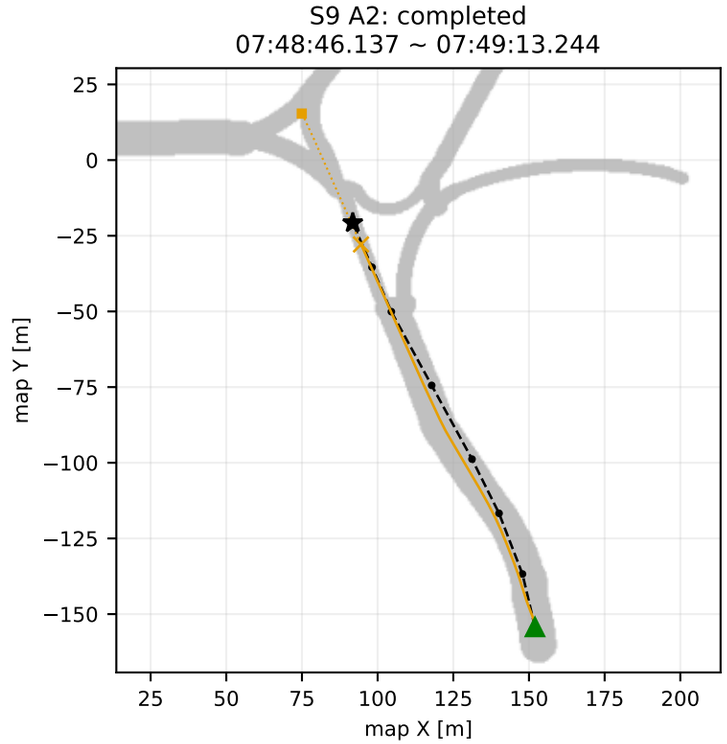


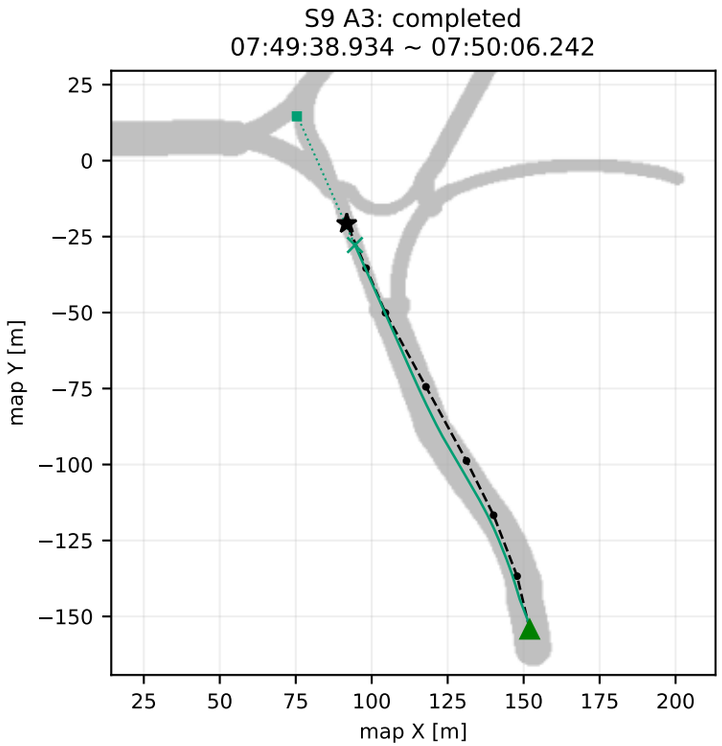


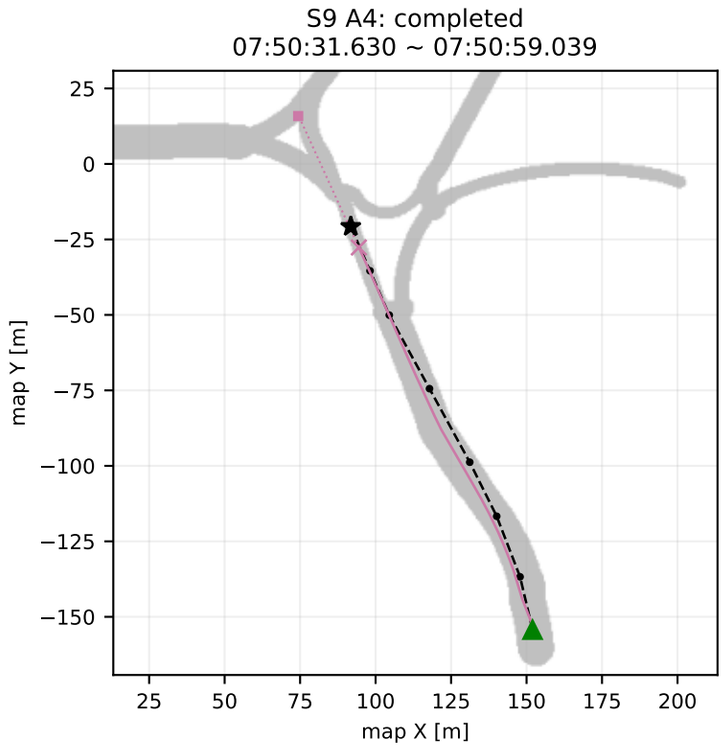
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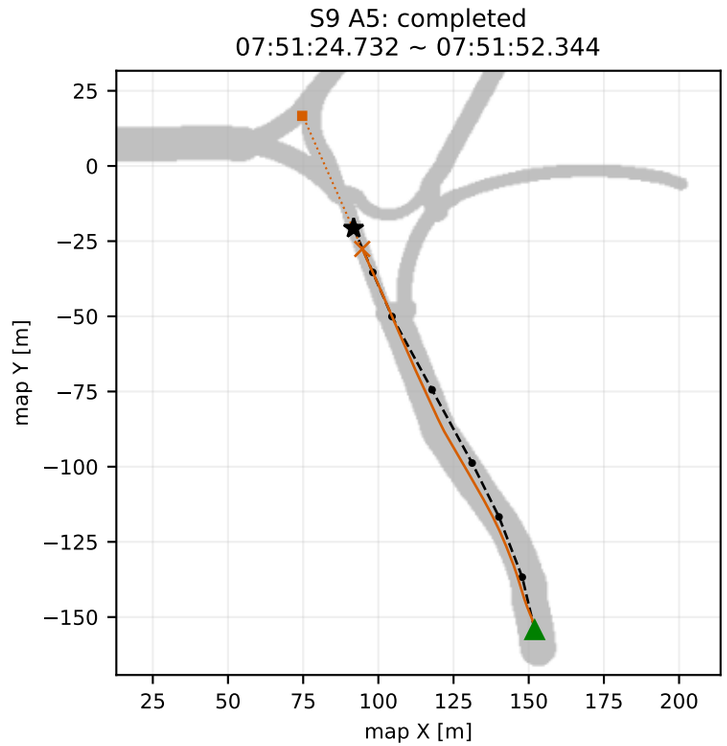




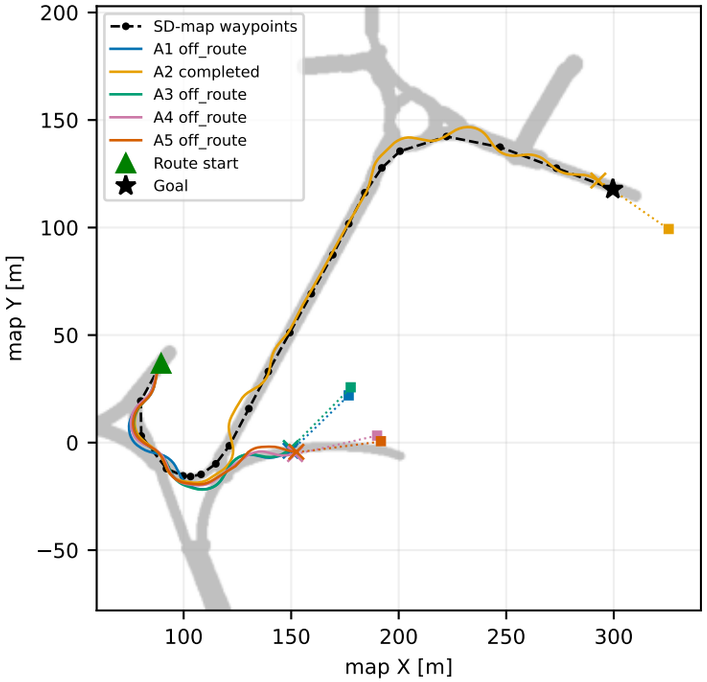


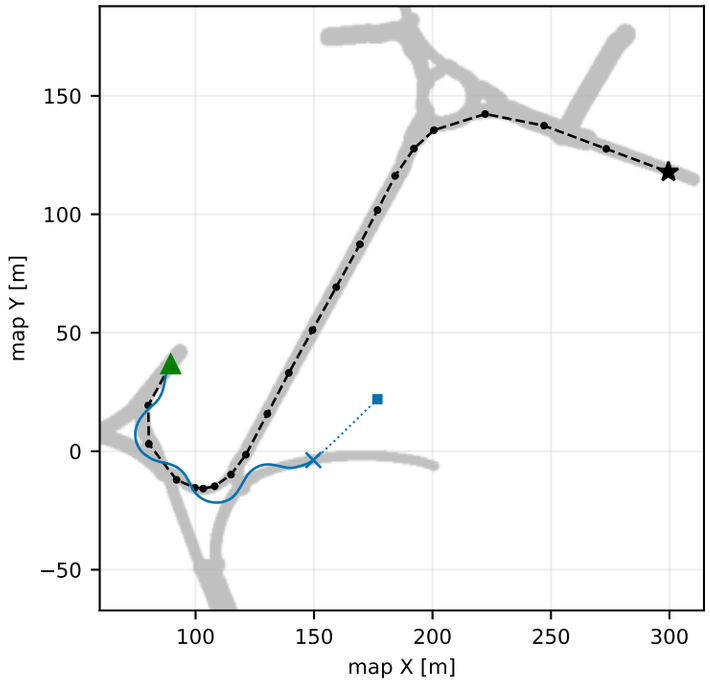


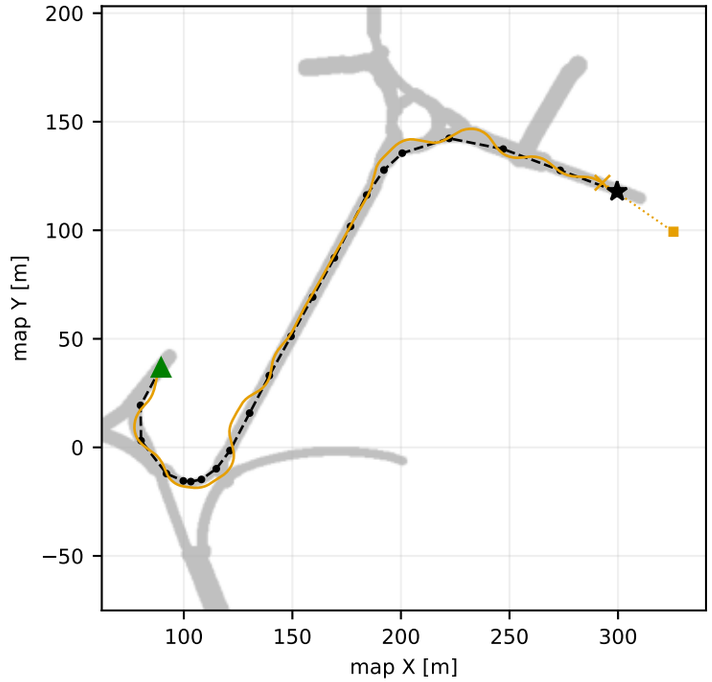


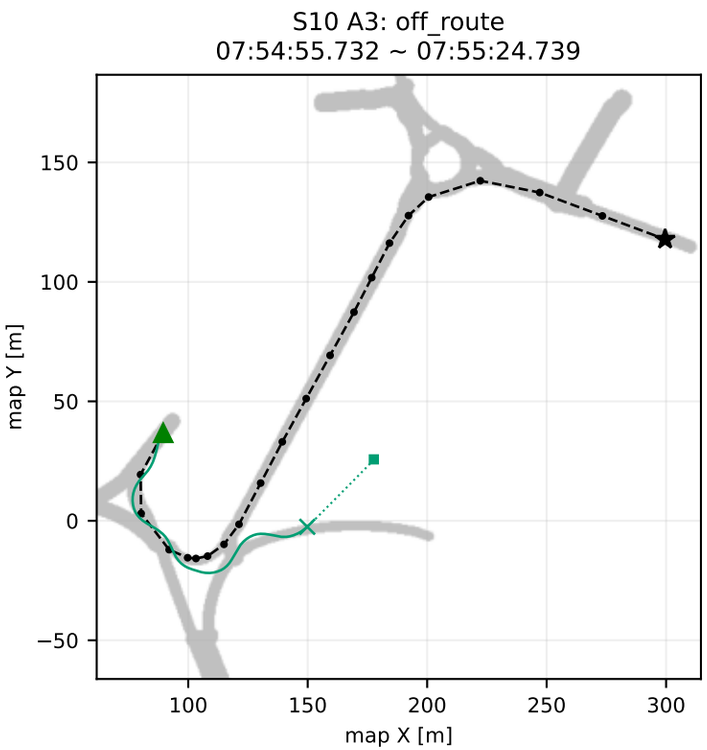


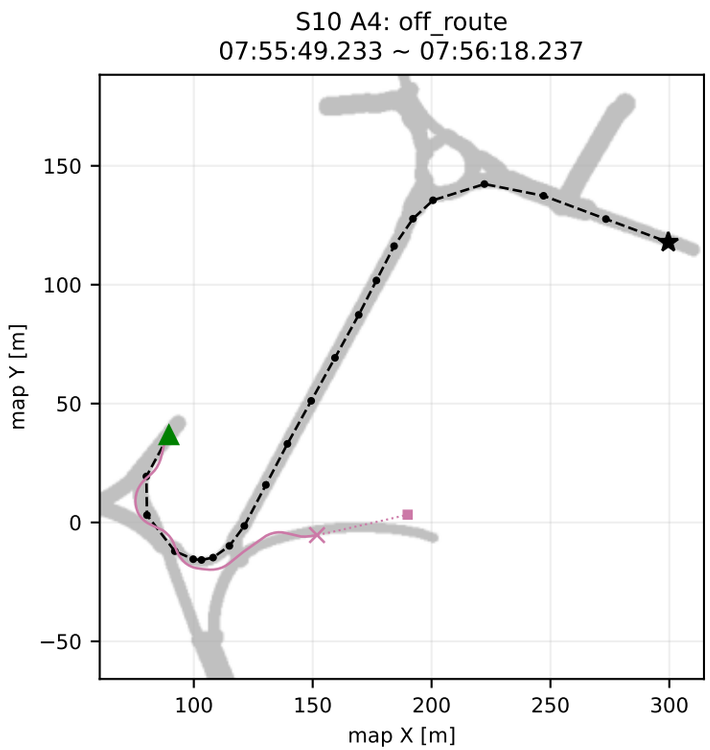
20261009_070714_all_d8c3cf | solid=running, dotted=coast; x=outcome, square=stop | design road backdrop

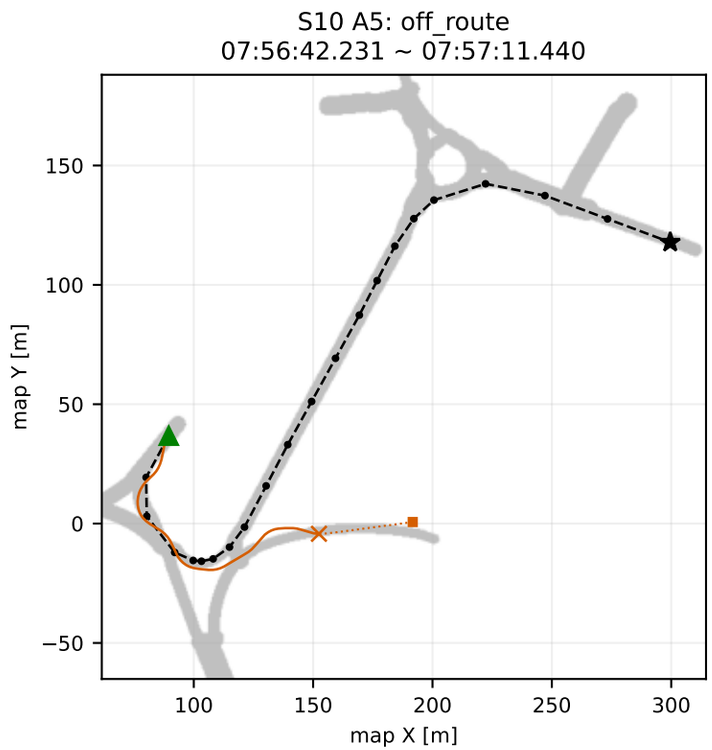




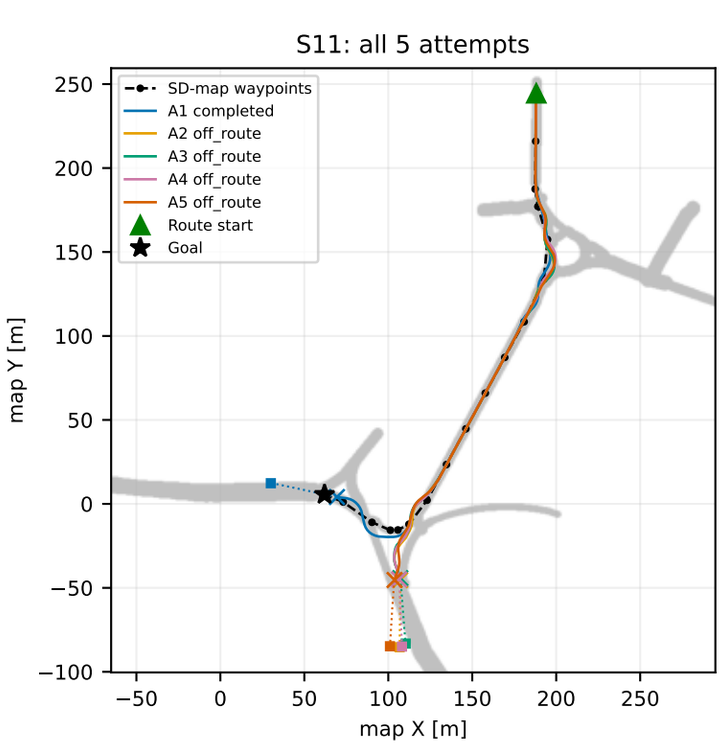


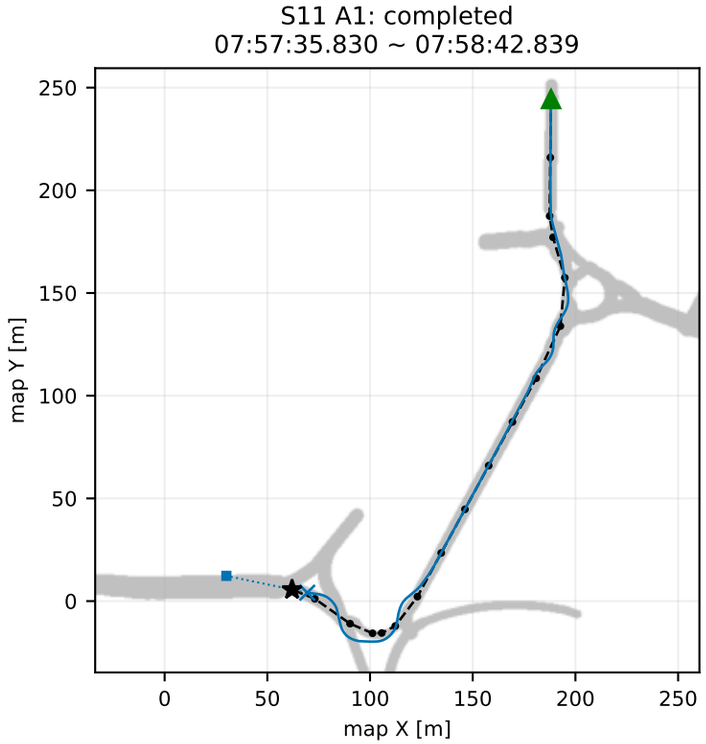


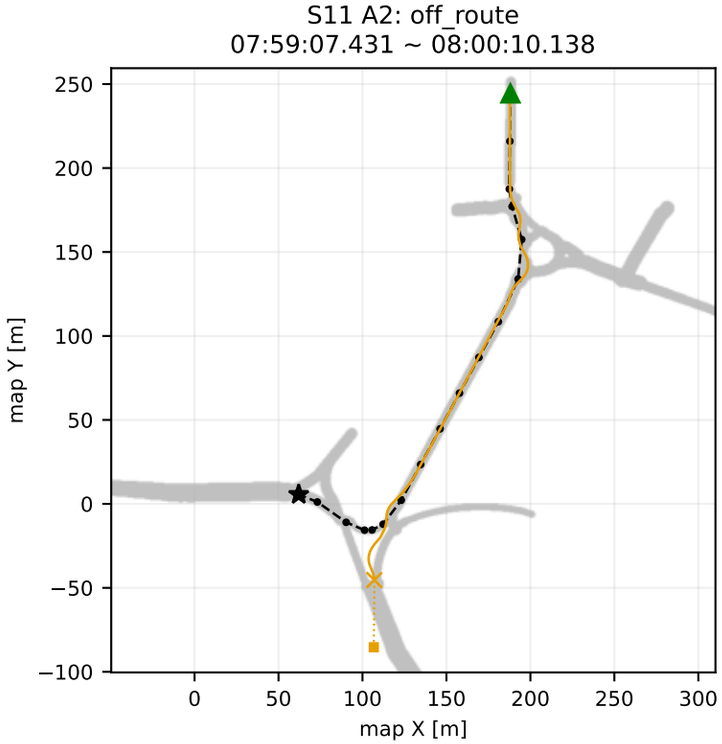


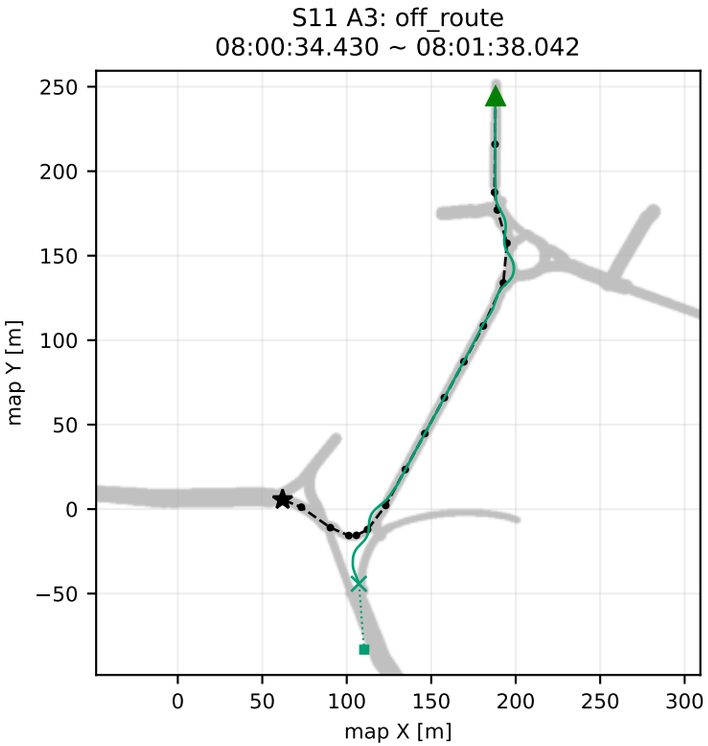


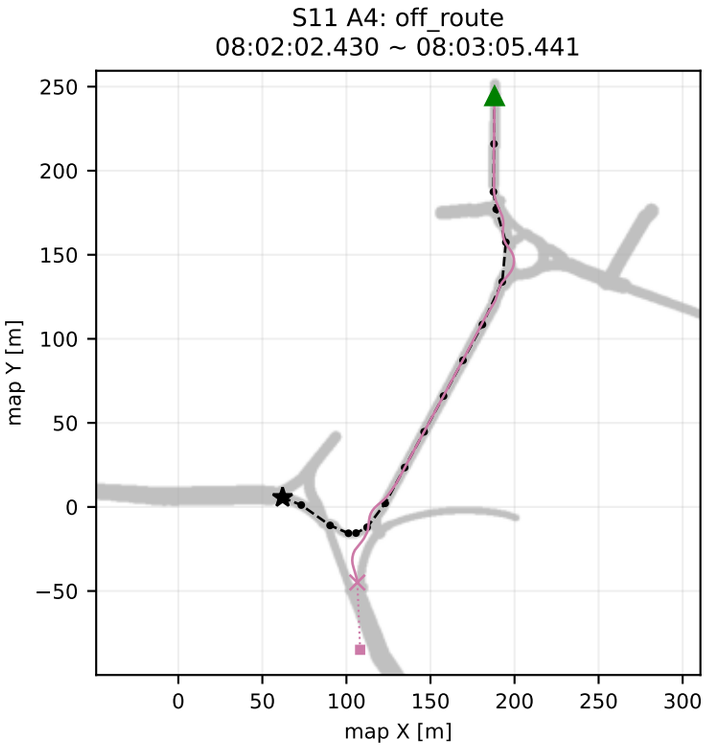
20261009_070714_all_d8c3cf | solid=running, dotted=coast; x=outcome, square=stop | design road backdrop

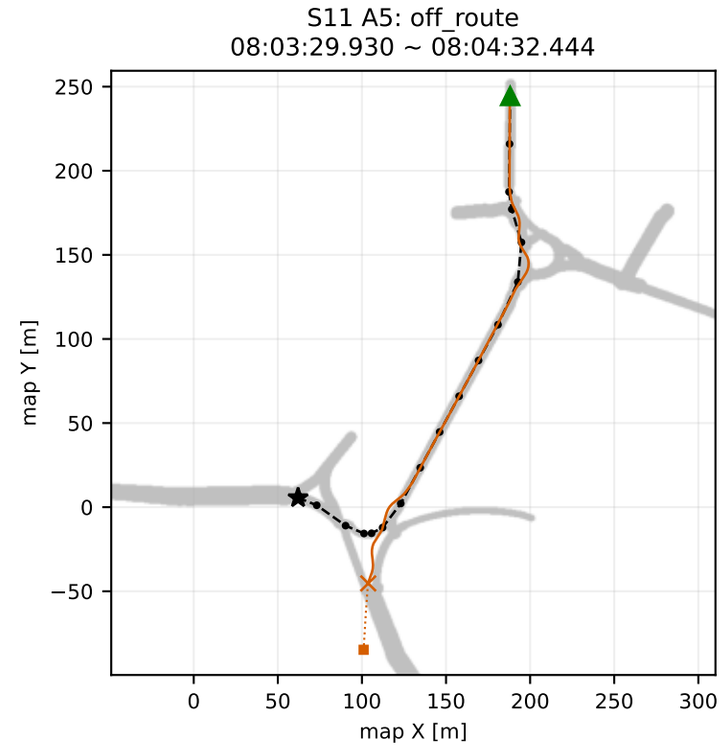




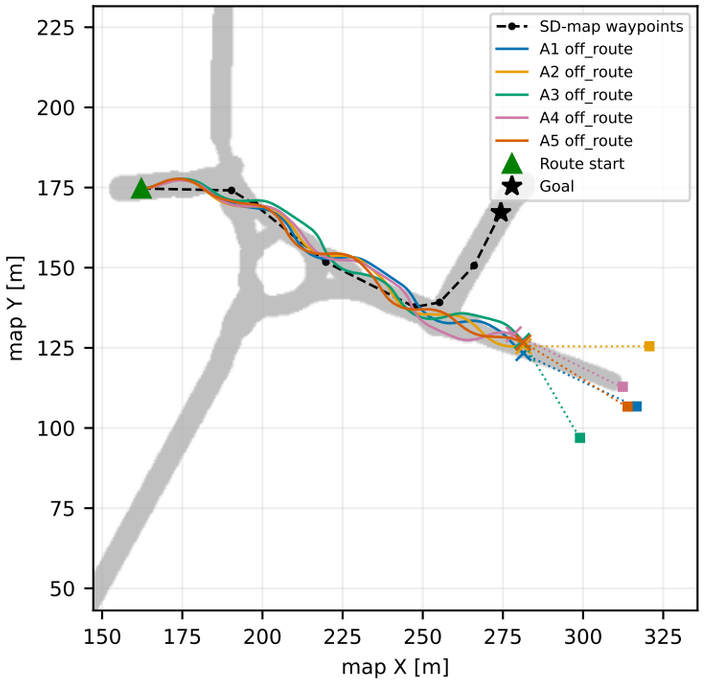


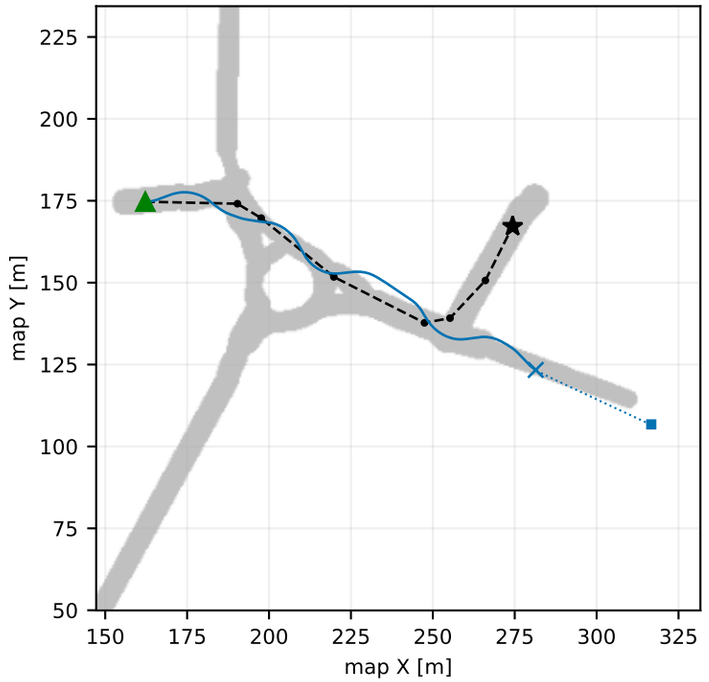


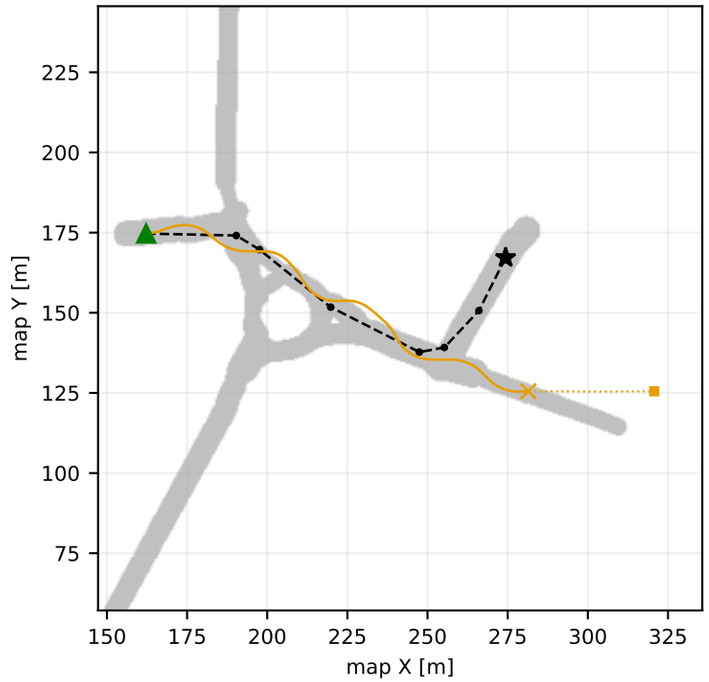


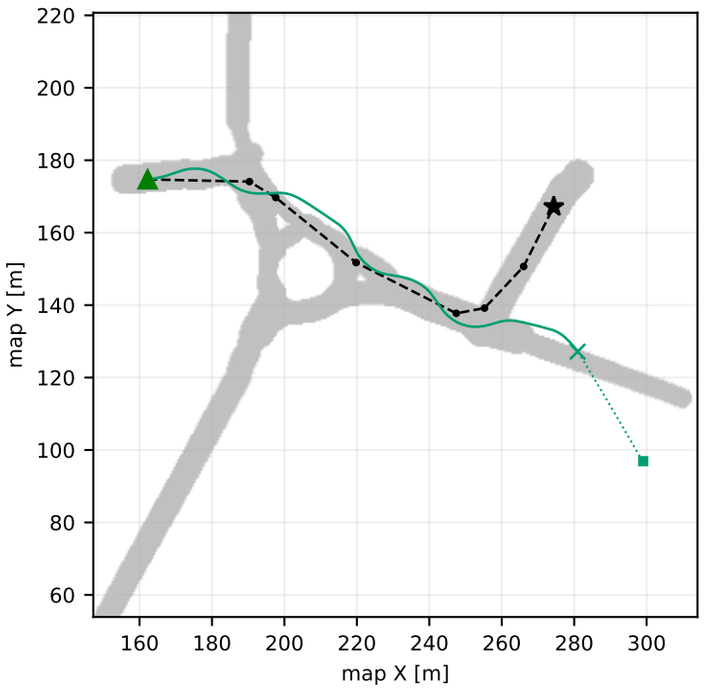


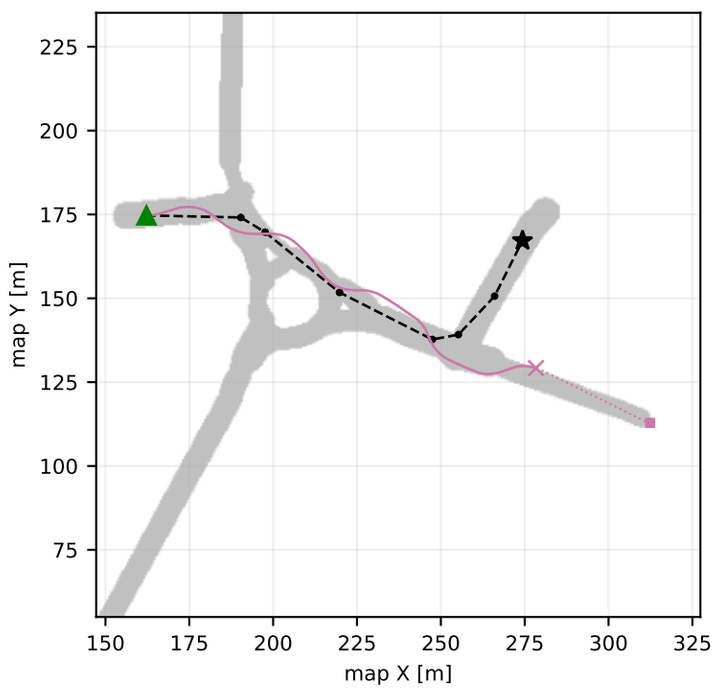
20261009_070714_all_d8c3cf | solid=running, dotted=coast; x=outcome, square=stop | design road backdrop

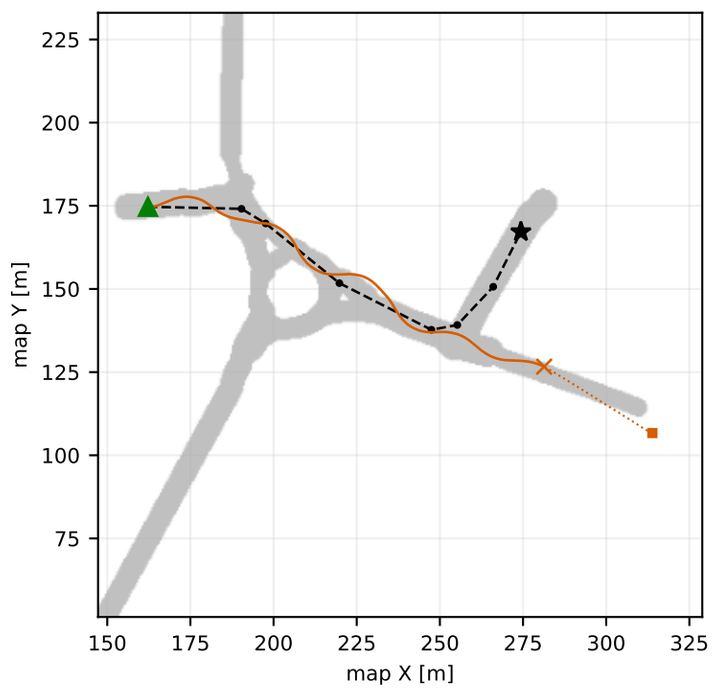


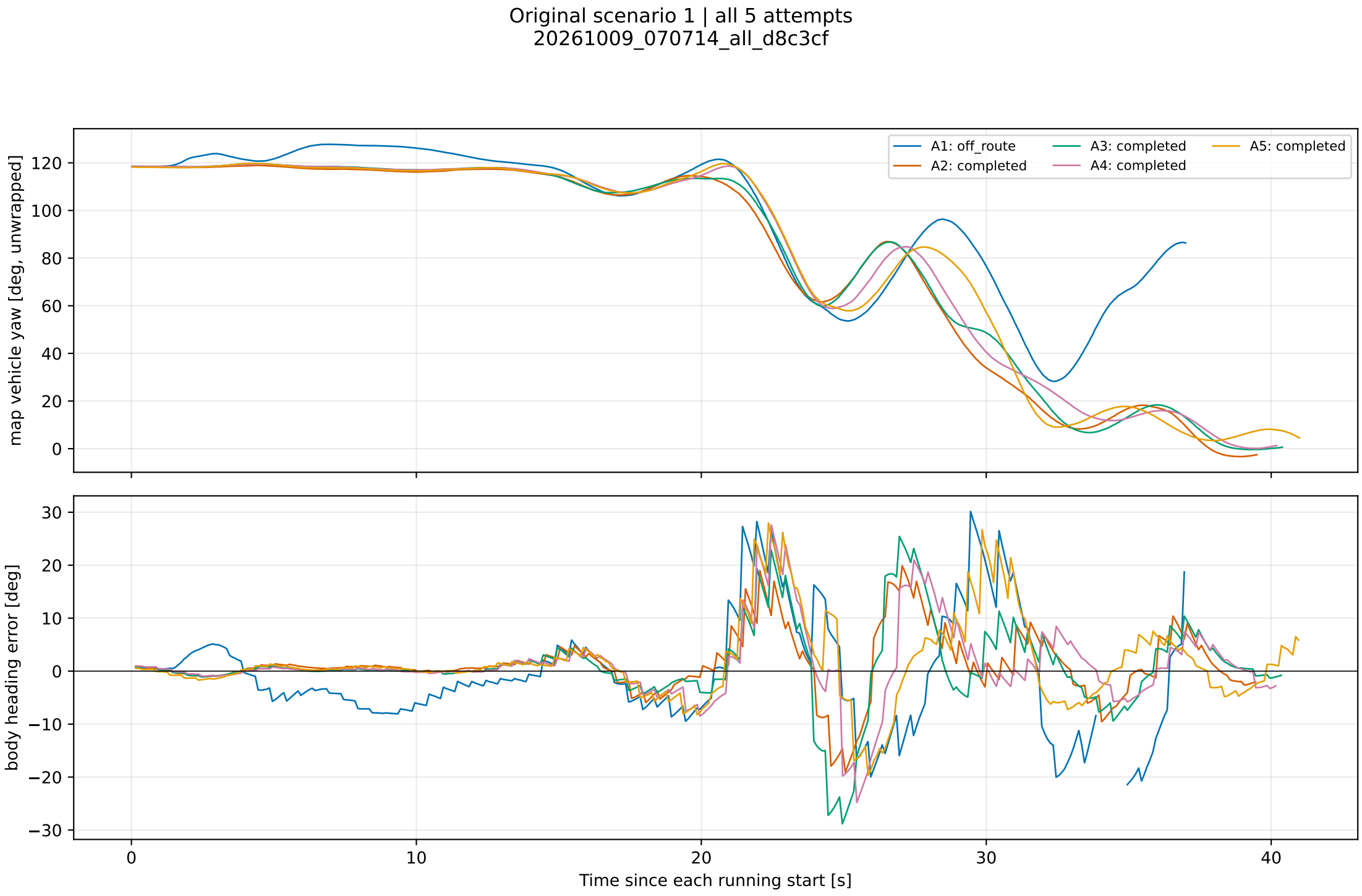


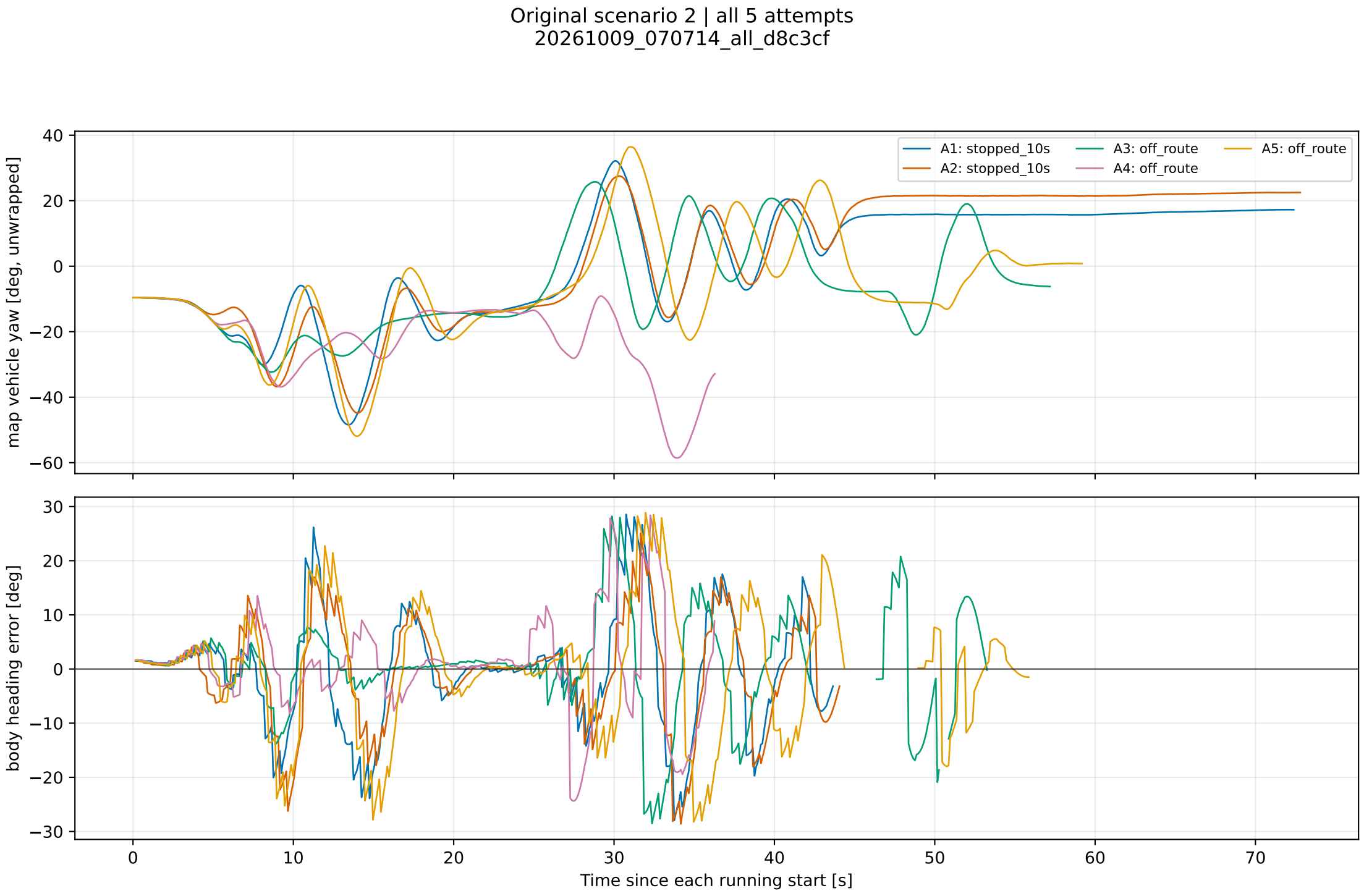


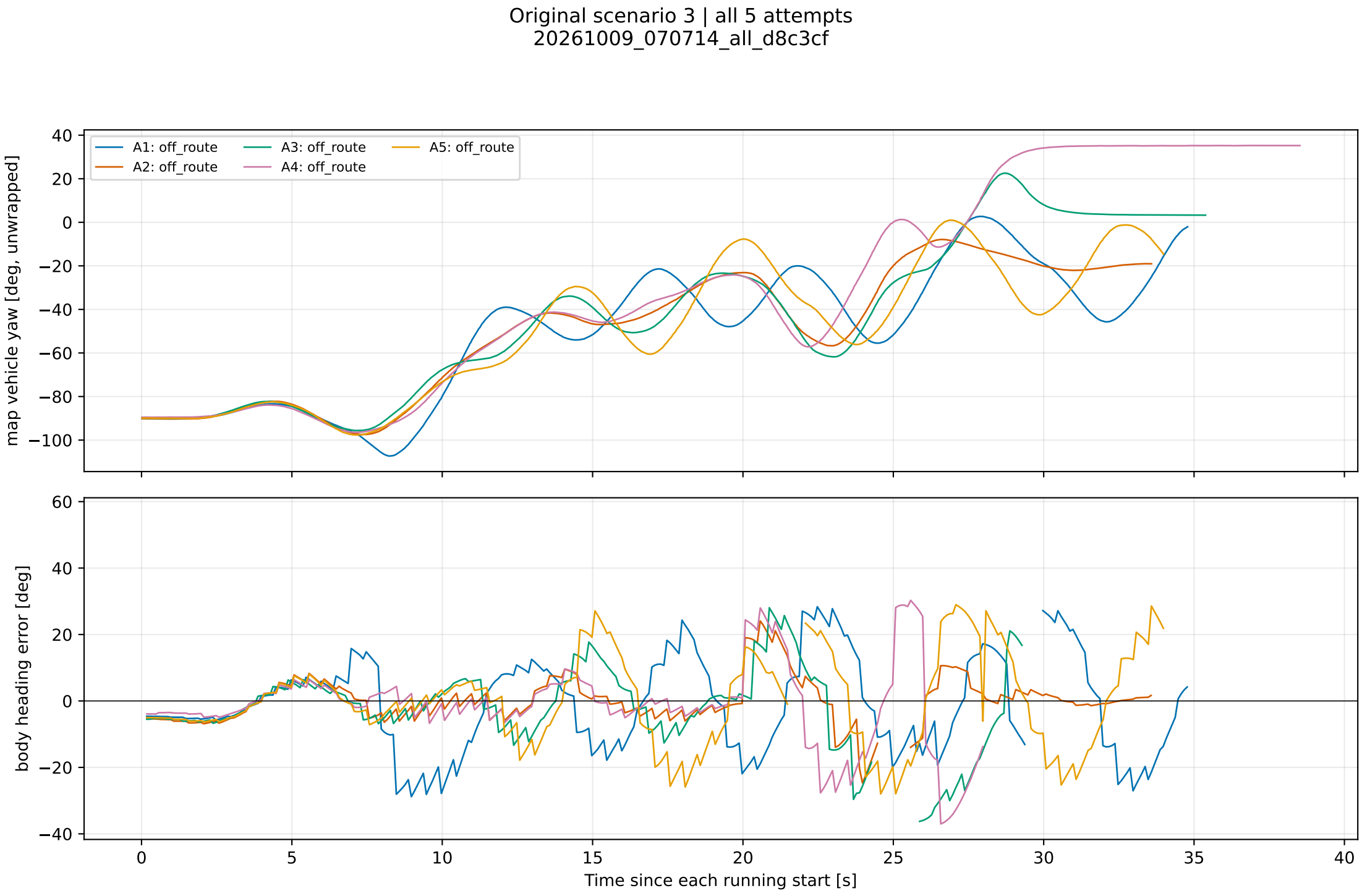




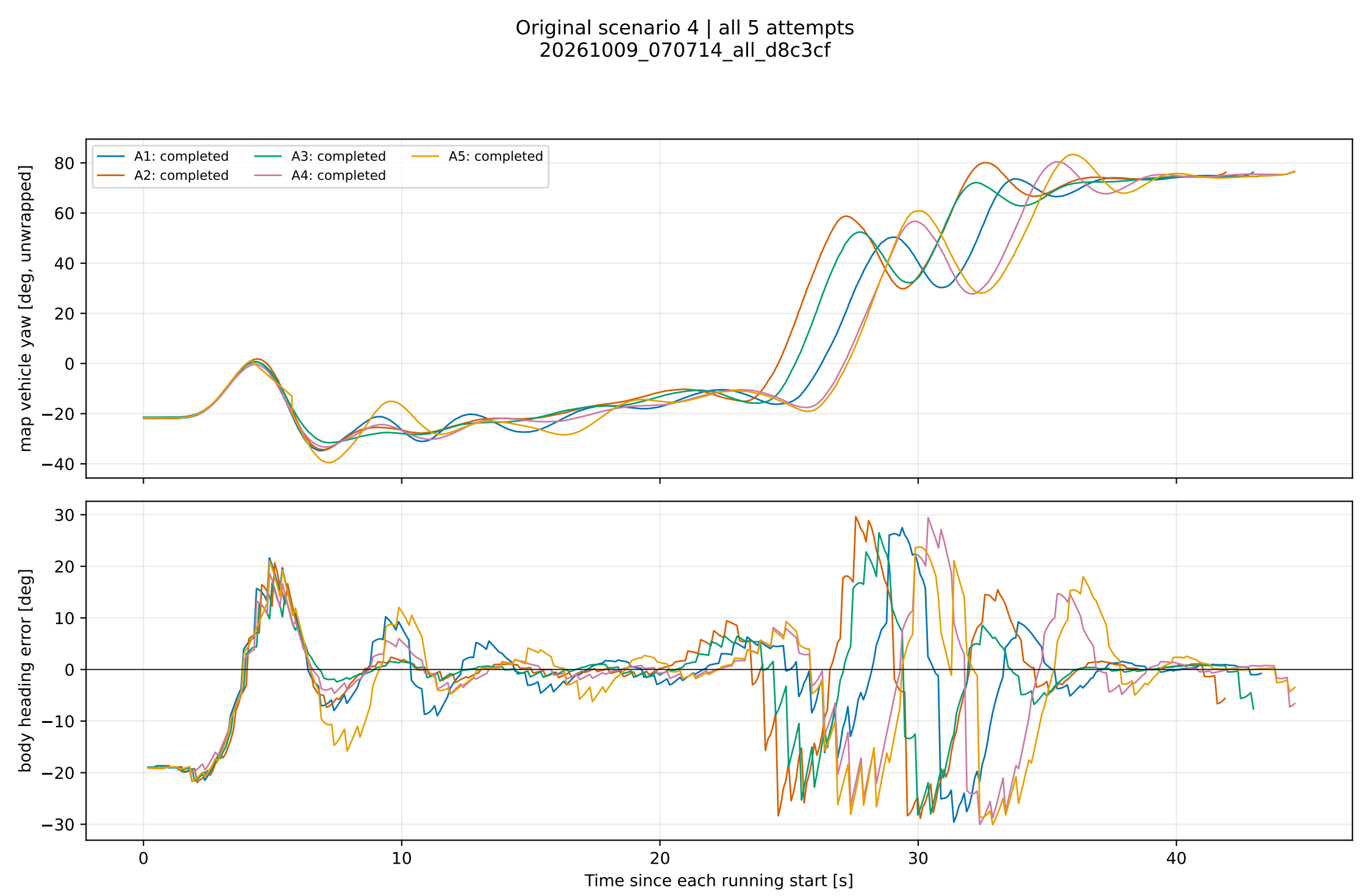



Original scenario 1 | all 5 attempts
20261009_070714_all_d8c3cf
A1: off_route
A2: completed
A3: completed
A4: completed
A5: completed
map vehicle yaw [deg, unwrapped]
body heading error [deg]
Time since each running start [s]
Top: scalar odom orientation.z; unwrap separately per attempt. Bottom: lower reference-path error.
Error shown only with valid QP or error feedback in modes 0/1/2/5; gaps are excluded samples. Not EKF-SIH difference.

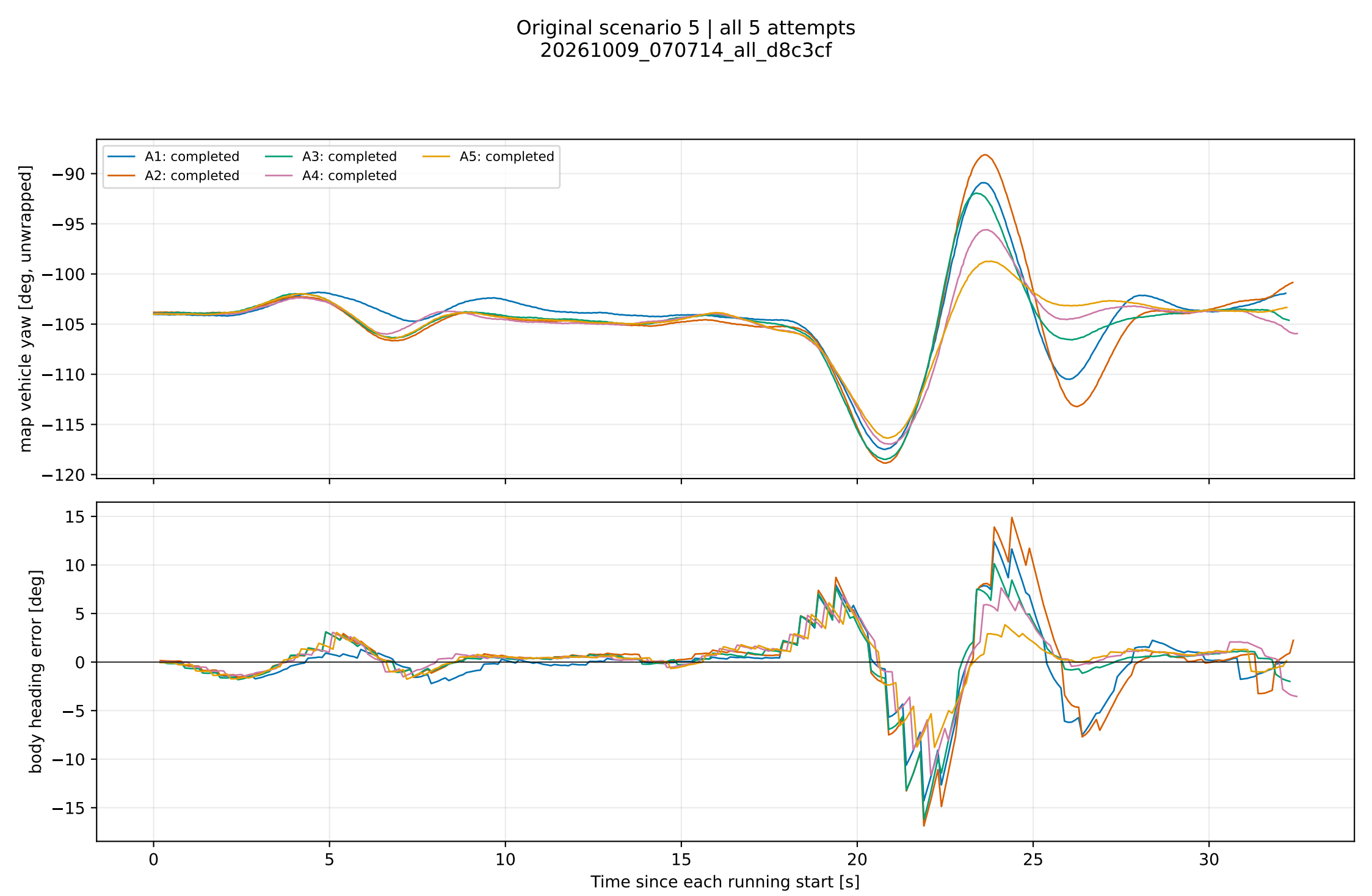

Original scenario 2 | all 5 attempts
20261009_070714_all_d8c3cf
A1: stopped_10s
A2: stopped_10s
A3: off_route
A4: off_route
A5: off_route
map vehicle yaw [deg, unwrapped]
body heading error [deg]
Time since each running start [s]
Top: scalar odom orientation.z; unwrap separately per attempt. Bottom: lower reference-path error.
Error shown only with valid QP or error feedback in modes 0/1/2/5; gaps are excluded samples. Not EKF-SIH difference.



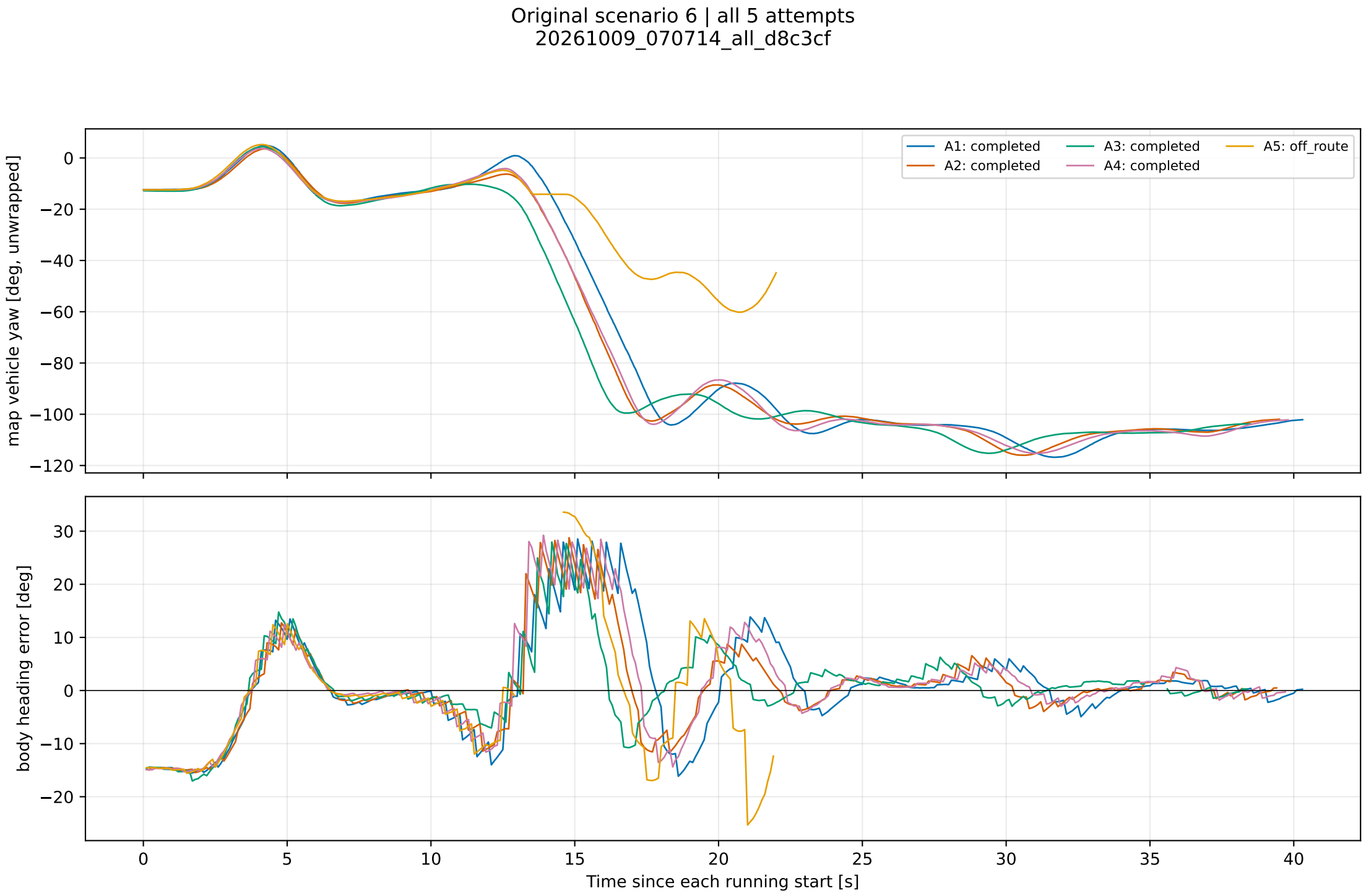
Top: scalar odom orientation.z; unwrap separately per attempt. Bottom: lower reference-path error.
Error shown only with valid QP or error feedback in modes 0/1/2/5; gaps are excluded samples. Not EKF-SIH difference.



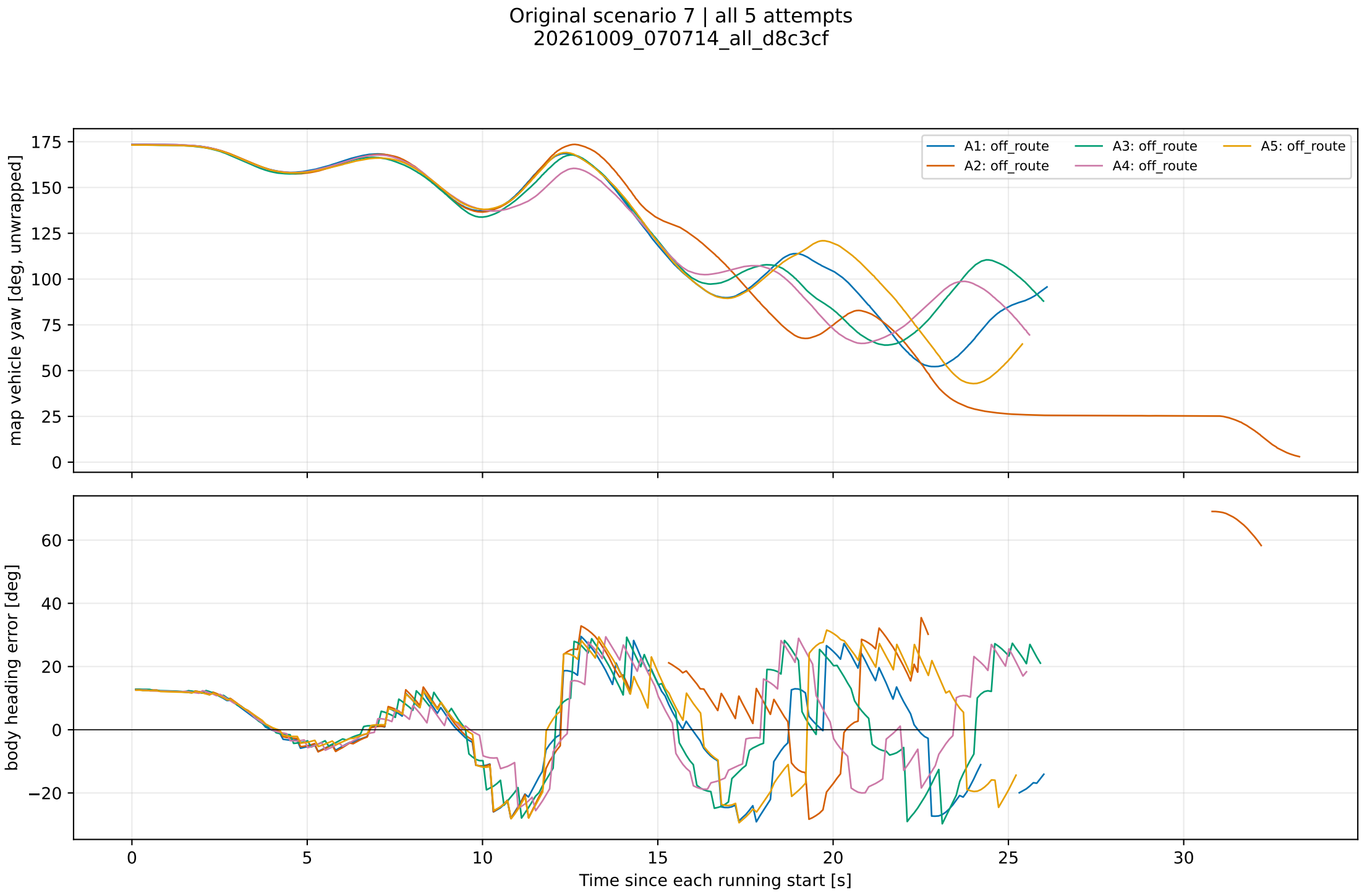
Top: scalar odom orientation.z; unwrap separately per attempt. Bottom: lower reference-path error.
Error shown only with valid QP or error feedback in modes 0/1/2/5; gaps are excluded samples. Not EKF-SIH difference.

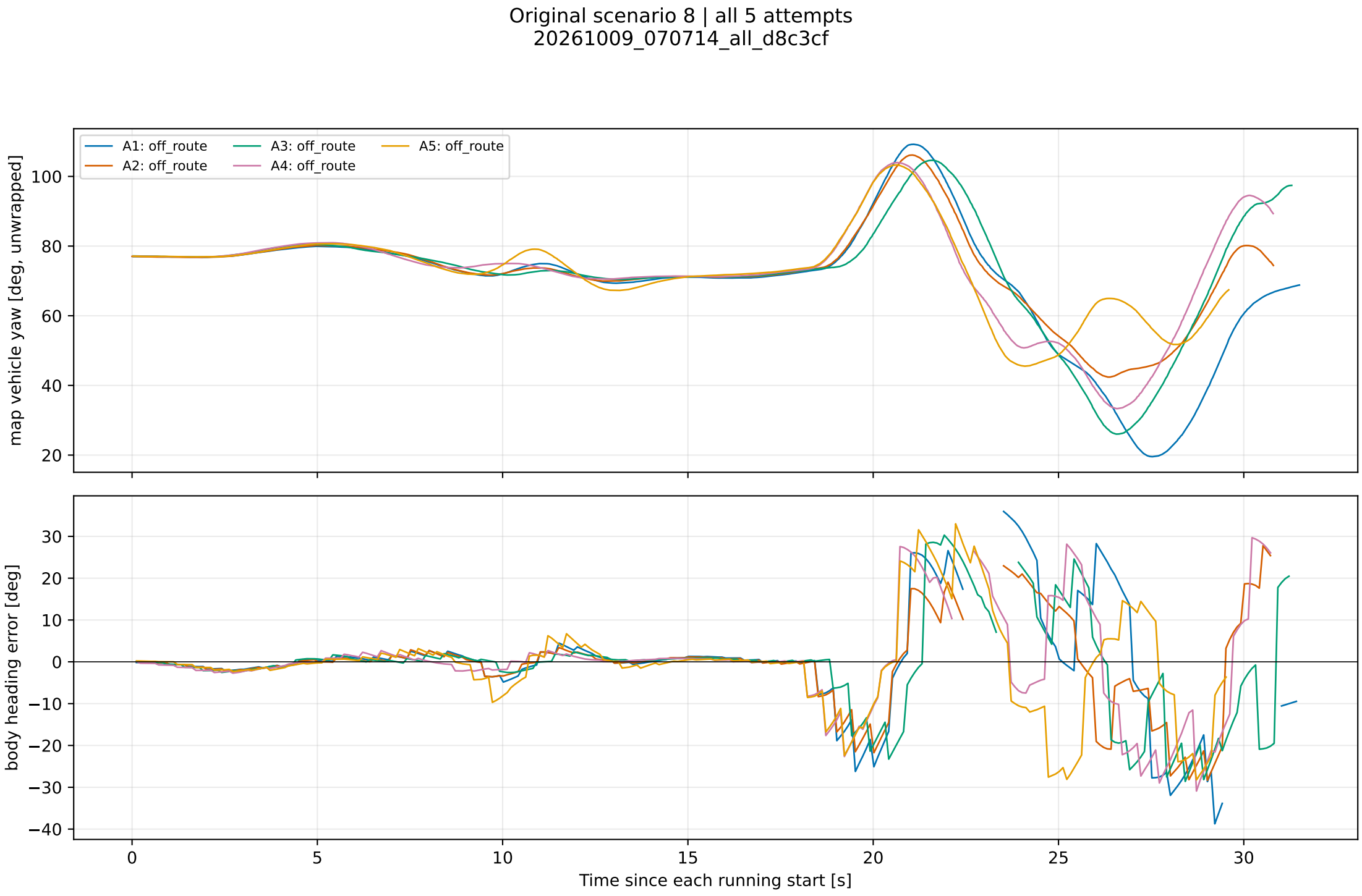


Top: scalar odom orientation.z; unwrap separately per attempt. Bottom: lower reference-path error.
Error shown only with valid QP or error feedback in modes 0/1/2/5; gaps are excluded samples. Not EKF-SIH difference.

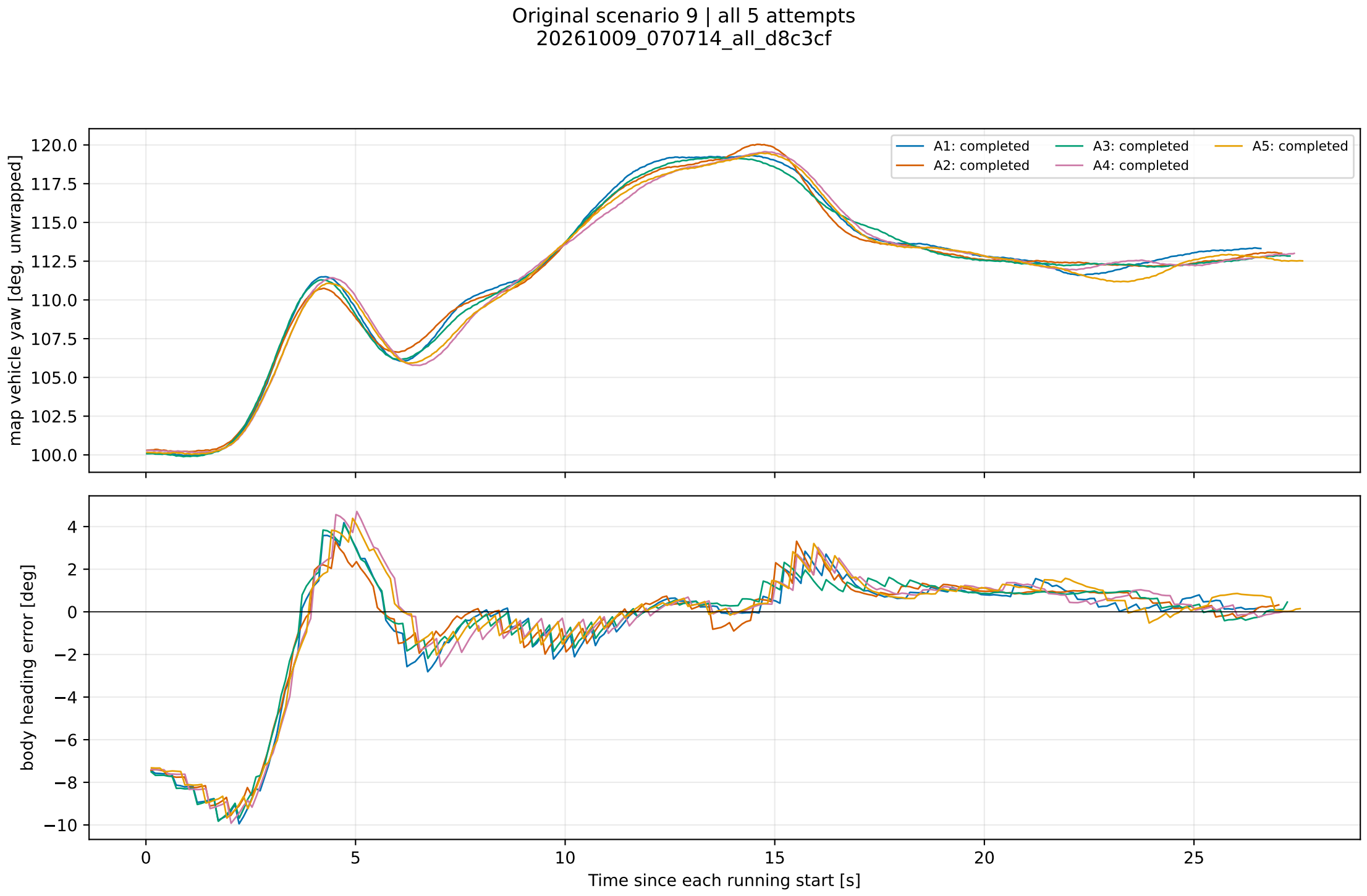


Top: scalar odom orientation.z; unwrap separately per attempt. Bottom: lower reference-path error.
Error shown only with valid QP or error feedback in modes 0/1/2/5; gaps are excluded samples. Not EKF-SIH difference.

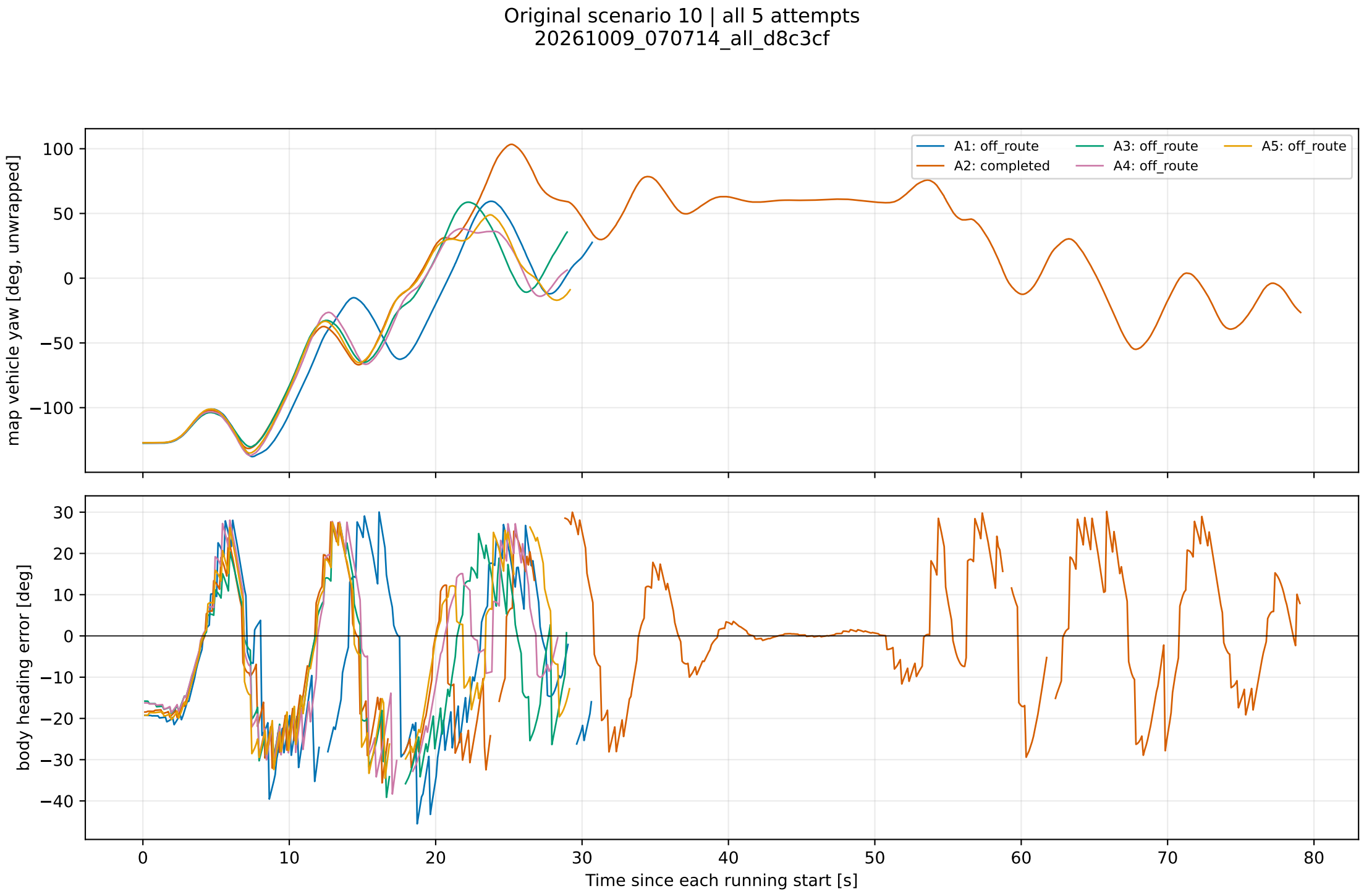

Original scenario 7 | all 5 attempts
20261009_070714_all_d8c3cf
A1: off_route
A2: off_route
A3: off_route
A4: off_route
A5: off_route
map vehicle yaw [deg, unwrapped]
body heading error [deg]
Time since each running start [s]
Top: scalar odom orientation.z; unwrap separately per attempt. Bottom: lower reference-path error.
Error shown only with valid QP or error feedback in modes 0/1/2/5; gaps are excluded samples. Not EKF-SIH difference.



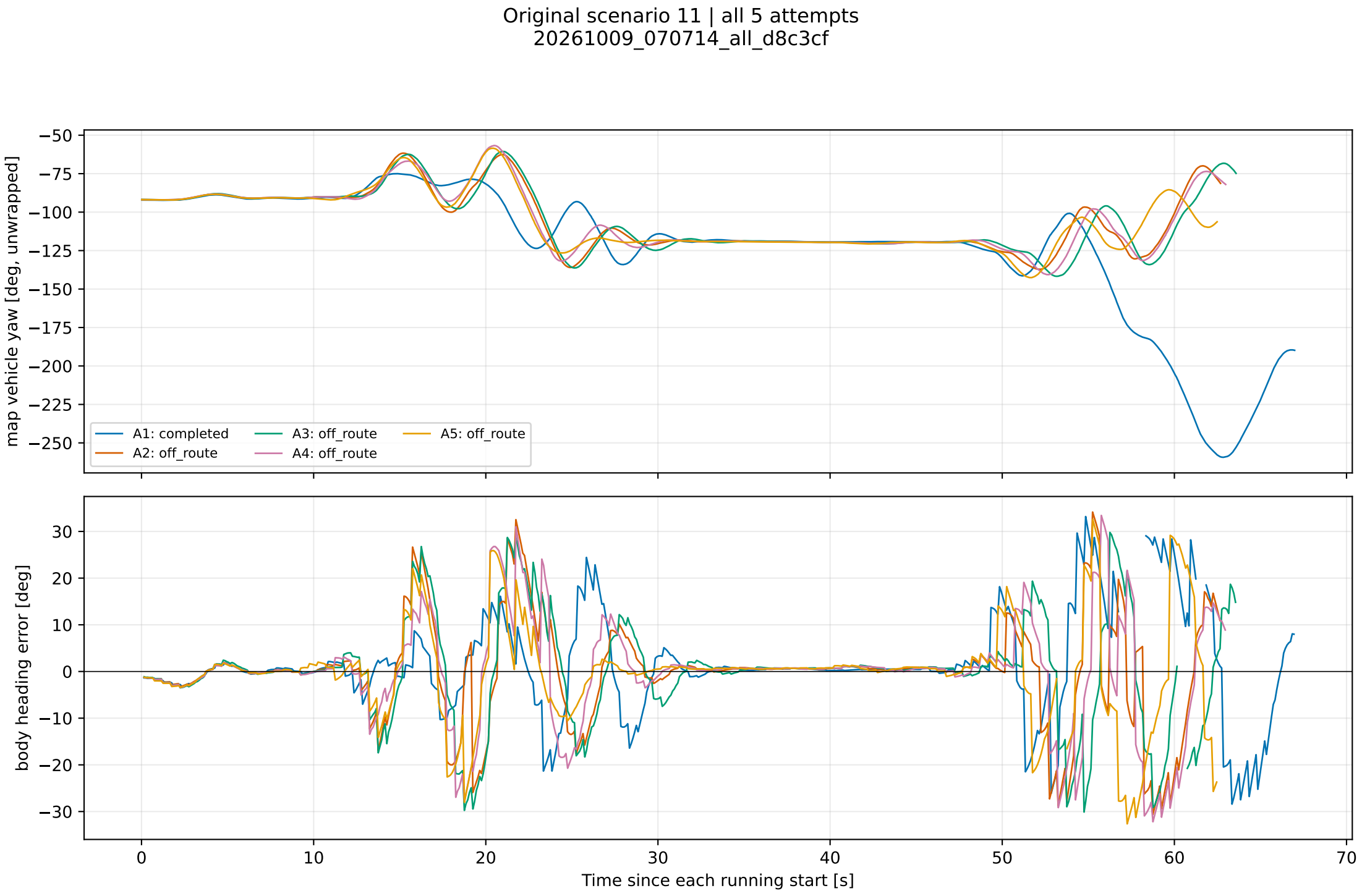
Top: scalar odom orientation.z; unwrap separately per attempt. Bottom: lower reference-path error.
Error shown only with valid QP or error feedback in modes 0/1/2/5; gaps are excluded samples. Not EKF-SIH difference.



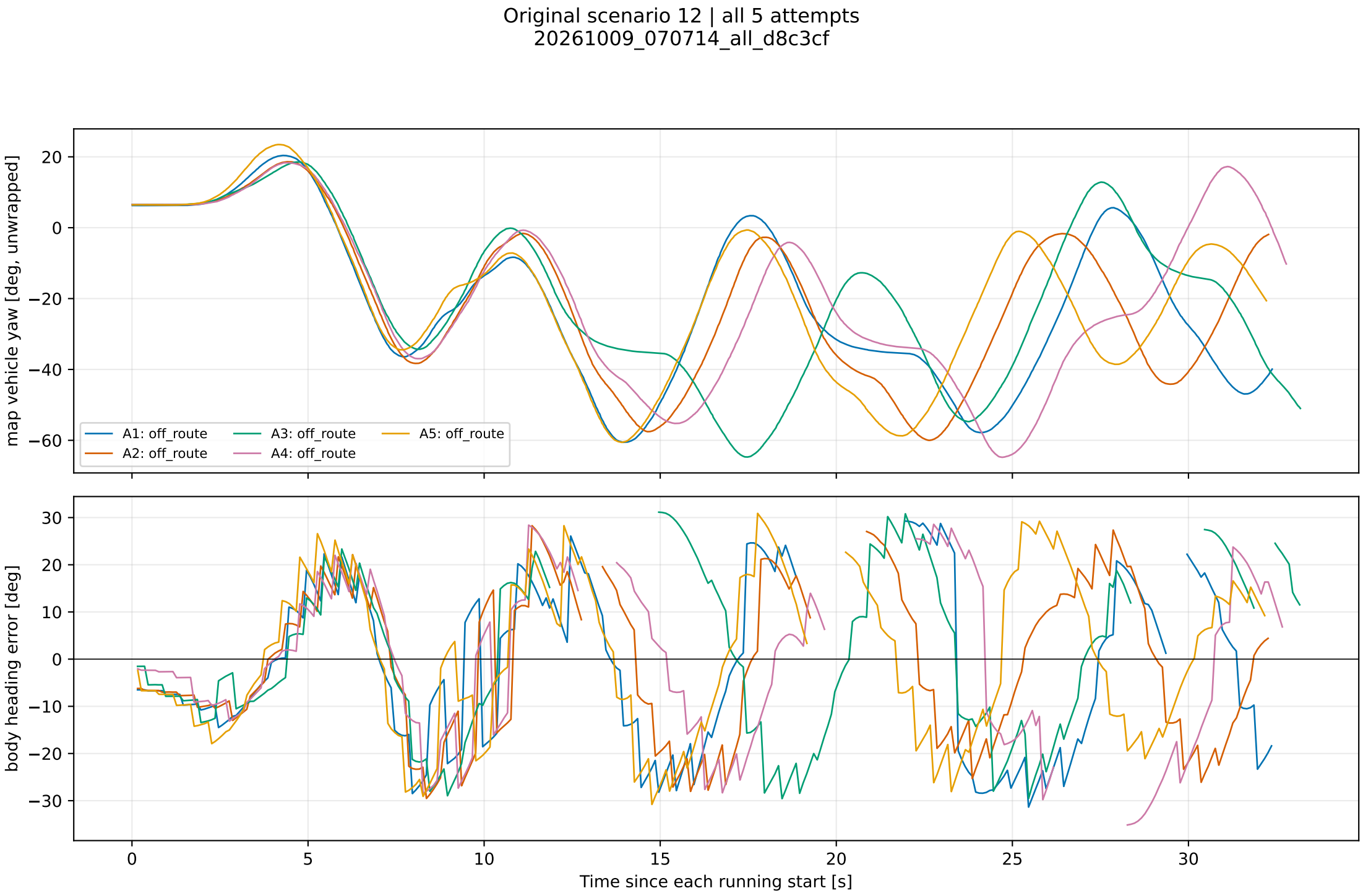
Top: scalar odom orientation.z; unwrap separately per attempt. Bottom: lower reference-path error.
Error shown only with valid QP or error feedback in modes 0/1/2/5; gaps are excluded samples. Not EKF-SIH difference.



Top: scalar odom orientation.z; unwrap separately per attempt. Bottom: lower reference-path error.
Error shown only with valid QP or error feedback in modes 0/1/2/5; gaps are excluded samples. Not EKF-SIH difference.



Top: scalar odom orientation.z; unwrap separately per attempt. Bottom: lower reference-path error.
Error shown only with valid QP or error feedback in modes 0/1/2/5; gaps are excluded samples. Not EKF-SIH difference.



Top: scalar odom orientation.z; unwrap separately per attempt. Bottom: lower reference-path error.
Error shown only with valid QP or error feedback in modes 0/1/2/5; gaps are excluded samples. Not EKF-SIH difference.